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MetaDex Scanner Administrator Guide

Informatica Enterprise Data Catalog MetaDex Scanner Administrator Guide
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Preface

Use the *MetaDex Scanner Administrator Guide* to learn how to administer MetaDex scanners, repository, and server. You can configure sources of metadata extraction, run and manage metadata jobs, configure the MetaDex server. Learn how to use the MetaDex tool to perform administrative tasks required for metadata scanning.

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CHAPTER 1

MetaDex administration overview

This chapter includes the following topics:

- Administration, 10
- MetaDex scanner process, 12
- Connection registry, 12
- DDL statements, 13
- CSV files, 14
- Custom stored procedures, 18
- Managing file definitions, 19
- Graph properties, 20

Administration

With MetaDex scanners, you can extend the default metadata import capabilities of a data catalog. You can analyze various types of data integration applications and gain business context by viewing detailed, end-to-end data lineage information about metadata in an enterprise.

Analyzing metadata and generating proper data lineage requires an execution environment that meets certain criteria. For example, for the MetaDex server to process SQL statements and infer lineage, you must define the connection registry. The connection registry is where the MetaDex server finds definitions of database objects to be used during the parsing process. A database object can be a table, a column, a synonym, or a database link.

Similarly, when analyzing ETL extracts, provide information about parameters and definitions of the database connections used by the ETL processes.

As a data administrator, use the MetaDex tool to perform administrative tasks. For example, you can manage permissions, manage files, or add and configure external tools.

Example

You work as a data analyst in a retail corporation that has outlets across the country. You are asked to analyze data sources and calculations that are necessary for price calculation in predictive analysis reports.

You ask the Enterprise Data Catalog admin team to scan all the data sources for the "Product Sales Data Report" in Microsoft Power BI.

The Enterprise Data Catalog admin team coordinates with different teams and scans all relevant data sources such as ADLS GEN2, Azure Synapse Analytics Stored Procedures, Azure Synapse Analytics SQL

Pools, Databricks Notebooks python code, Azure Data Factory and Microsoft Power BI Reports with the use of the MetaDex tool.

They inform you that lineage is automatically built by scanning all these data sources and it is ready for your analysis. Now you have all the relevant information needed for your analysis in the Catalog tool. You can go to the Catalog tool and find the sources of data for your Product Sales Data Report.

Note: A Product Sales Data Report contains analytical data that comes from multiple systems. Data scientists perform predictive analysis on those data sources using Python and Stored Procedures.

The following table lists the tasks that you must complete and the steps that you must perform in Enterprise Data Catalog to accomplish the task:

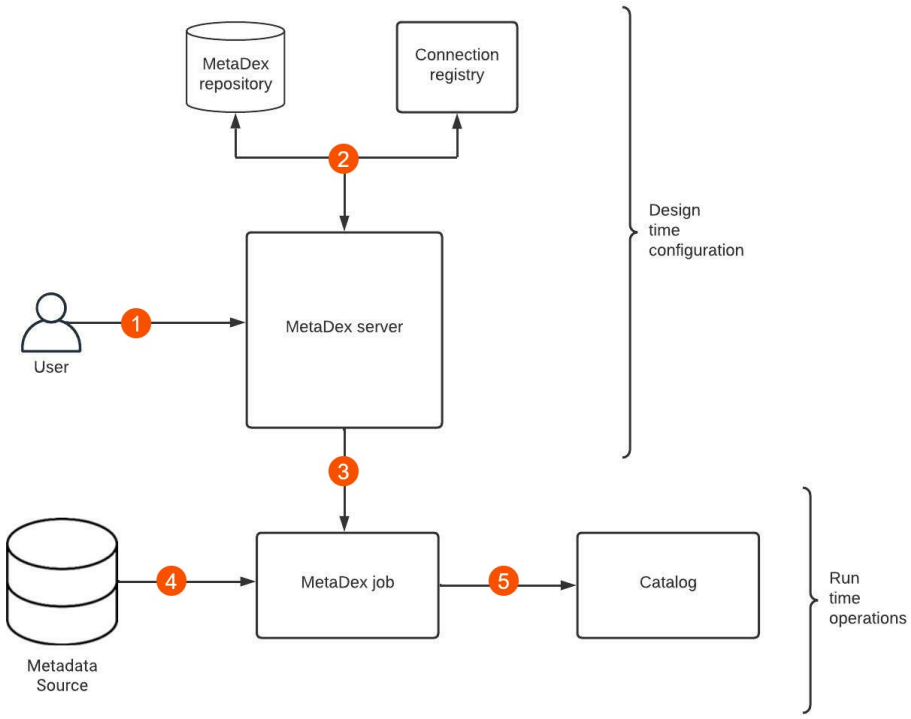
Tasks	Steps Using Enterprise Data Catalog
In the Catalog tool, search for the Product Sales Data Report .	To search for the report, use the search feature available in the Catalog tool.
Discover the sources that contribute data to the Product Sales Data Report.	To view the details of the report and identify the data sources of the report, perform the following steps: 1. Go to **Lineage and Impact** > **Report and Expand Lineage**. 2. Click the "Plus" icon to expand all systems. 3. Click the " Show Transformation logic in lineage" icon. **Note:** Orange dots next to asset names indicate these assets have been transformed.
Verify the sources and transformation logic applied to assets.	See and analyze the calculations and rules used in Stored Procedures, Databricks Notebooks and Azure Data Factory Pipelines at the table and column level.
View the source code in the system attributes.	To see the source code of Stored Procedures and Databricks Notebooks, use the System Attributes Properties option.
View detailed lineage using MetaDex Visualization.	To verify the detailed lineage in the MetaDex tool, use the MetaDex Visualization links.

For more information, see "Business Examples" in the *Enterprise Data Catalog User Guide*.

MetaDex scanner process

Use MetaDex scanners to extract metadata from external metadata sources, and then load that metadata into the catalog. Enterprise Data Catalog users, such as data stewards or analysts can use the extracted metadata.

The following image shows the process for MetaDex scanners:



1. You connect with the MetaDex server to create a configuration. The MetaDex server is service that hosts the web-based MetaDex tool. The MetaDex server allows you to manage scanner configurations and trigger MetaDex scanner jobs. You can also use it to work with logs to identify and fix problems that arise during the scanning process.
2. The MetaDex database repository and the filesystem connection registry store the configurations you created.
3. When you start a scan, the MetaDex server creates a job that performs the scan operation.
4. The MetaDex job reads metadata from the source.
5. The MetaDex job sends the extracted metadata to the catalog.

Connection registry

The connection registry stores database connections.

The connection registry is organized in a two-lever hierarchy. The first level is the host, and the second is the database level.

You can distinguish the following types of connection registry entries:

- Database connection parameters defined in the db.properties file.
- A set of CSV files that describe the tables in the repository.
- SQL files that contain `create table` statements.
- The Files.csv file that describes the missing column names. You place this type of entry on the host level.

Note: You place the first three types in the database level. These types require a db.properties file. This file must contain information about the method type and vendor of the database. A db.properties file can also contain additional information, such as settings required for a certain database or dblinks that allow you to connect to other databases.

DDL statements

Data Definition Language (DDL) statements are a subset of SQL statements that change the structure of the database schema.

The following example presents basic `CREATE TABLE` DDL statements which can be provided for one or more files:

SQL DDL

```
CREATE TABLE CSPL.INPUT_TAB2 (ID DECIMAL(9,0) NOT NULL
, COL1 CHAR(30) NOT NULL
, COL2 CHAR(30) NOT NULL
, COL3 CHAR(30) NOT NULL
, COL4 DECIMAL(9,0) NOT NULL
);

CREATE TABLE CSPL.INPUT_TAB3 (ID DECIMAL(9,0) NOT NULL
, COLUMN1 CHAR(30) NOT NULL
, COLUMN2 CHAR(30) NOT NULL
, COLUMN3 CHAR(30) NOT NULL
, COLUMN4 DECIMAL(9,0) NOT NULL
);
```

The table name comes from the `CREATE TABLE` statement. The identifying prefixes are built from the file path in the repository.

Oracle Vendor

```
vendor=ORACLE
repoType=DDL
```

To use the DDL method, you must create a directory structure for the repository. The DDL repository uses the following structure: `HOST/DATABASE/SCHEMA`.

The `db.properties` file is in the database level, and the DDLs are in the schema level.

CSV files

The connection registry uses CSV files to copy the metadata of the data sources.

You can find scripts and instructions to generate CSV files in one of the following ways:

- Go to the directory where you extracted the installer files and navigate to the following subdirectory: `utils/dbToCatalogRepo`

  Note: The CSV files must have the same names as catalog tables.
- Use the Metadata Extractor utility and a generated ZIP file containing the CSV files. The MetaDex server supports multiple ZIP files, for example, a single file for each schema. For more information about the Medatata Extractor utility, see "Metadata Extractor overview" on page 85.

Note: You can find additional ZIP files in the following directory: `utils/mdrepo`. For Oracle, you can use a file containing DUAL table-related entries instead of providing a SYS schema only when using a DUAL table.

View the database objects definitions for the following vendors:

- Oracle
- DB2
- Teradata
- Microsoft SQL
- Netezza
- SAS

db.properties for Oracle

```
vendor=ORACLE
repoType=CATALOG
```

Oracle table definition

You can provide table definitions in CSV files of the following form:

```
OWNER,TABLE_NAME,COLUMN_NAME,DATA_TYPE,COLUMN_ID
SYS,DUAL,DUMMY,VARCHAR2,42
SYS,SYSTEM_PRIVILEGE_MAP,PRIVILEGE,NUMBER,2
SYS,SYSTEM_PRIVILEGE_MAP,NAME,VARCHAR2,14
```

Oracle synonym definition

You can define Oracle synonyms in CSV files as in the following example:

```
OWNER,SYNONYM_NAME,TABLE_OWNER,TABLE_NAME,DB_LINK
PUBLIC,V$FAST_START_SERVERS,SYS,V_$FAST_START_SERVERS,
PUBLIC,V$FAST_START_TRANSACTIONS,SYS,V_$FAST_START_TRANSACTIONS,
PUBLIC,V$FILEMETRIC,SYS,V_$FILEMETRIC,
```

Oracle database link definition

You can define Oracle dblinks as in the following example:

```
OWNER,DB_LINK,USERNAME,HOST
PUBLIC,FINANCE.WORLD,FIN_SELECT,FINANCE
PUBLIC,CLAIMS.WORLD,CL_SELECT,CLAIMS
PUBLIC,CUSTOMERS.WORLD,CUS_SELECT,CUSTOMERS
```

When you use links, for every HOST, which is the last column in the `dblink` file, you must configure mapping from the HOST to a MetaDex repository database. Place the mapping in the `db.properties` file with the connection parameter in the following form:

```
HOST=TARGETHOST.TARGETDATABASE
```

Oracle database link-host mapping

The following example of `dbproperties` when a database link links the stgworld name to the STG database on ORACLE host:

```
vendor=ORACLE
repoType=CATALOG
connection=stgworld=ORACLE.STG
```

To specify more than one link, you can append a numeric ID to the connection parameters in the following example:

```
vendor=ORACLE
repoType=CATALOG
connection.0=stgworld=ORACLE.STG
connection.1=martworld=ORACLE.MART
```

Oracle columns definition

The following example shows the Oracle columns definition format:

```
"OWNER","TABLE_NAME","COLUMN_NAME","COLUMN_ID"
"SYS","DUAL","DUMMY","42"
"SYS","DBA_ALL_TABLES","PARTITIONED","54"
"SYS","DBA_ALL_TABLES","COMPRESSION","121"
```

DB2 table definition

The following example shows the SYSCAT.TABLES.csv format:

```
TABSCHEMA,TABNAME,TYPE,BASE_TABSCHEMA,BASE_TABNAME
 SYSIBM  ,SYSTABLES,T,NULL,NULL
SYSIBM  ,SYSCOLUMNS,T,NULL,NULL
SYSIBM  ,SYSINDEXES,T,NULL,NULL
```

To extract tables comments, include a REMARKS column in the CSV file.

DB2 column definition

```
TABSCHEMA,TABNAME,COLNAME,COLNO
SYSIBM,SYSTABLES,NAME42
SYSIBM,SYSTABLES,CREATOR,33
SYSIBM,SYSTABLES,TYPE,17
```

To extract table comments, include the following columns in the CSV file:

- SCALE
- TYPENAME
- LENGTH
- NULLS
- REMARKS

DB2 nickname definitions

The following example shows the SYSCAT.NICKNAMES.csv format:

```
REMOTE_TABLE,REMOTE_SCHEMA,SERVERNAME
MAGR_NAMES,MAGR,MYSAMPLE
```

DB2 useroptions definitions

The following example shows the SYSCAT.USEROPTIONS.csv format:

```
"AUTHID","AUTHIDTYPE","SERVERNAME","OPTION","SETTING"
"MAGR","U","MYSAMPLE","REMOTE_AUTHID","magr"
"MAGR","U","MYSAMPLE","REMOTE_PASSWORD","********"
```

DB2 views definitions

The following example shows the SYSCAT.VIEWS.csv format:

```
"VIEWSCHEMA","VIEWNAME","OWNER","TEXT","DEFINER"
"WOMA    ","V","WOMA    ","WOMA    ","create view v as (select * from
A)","WOMA    "
"TOBI","a_view","TOBI","TOBI","CREATE VIEW ""TOBI"".""a_view"" (A0, a1, a2)
AS select * from a","TOBI"
```

DB2 db.properties

Provide information about vendor and repository type:

```
vendor=DB2
repoType=CATALOG
```

Teradata db.properties file

Provide information about vendor and repository type:

```
vendor=TERADATA
repoType=CATALOG
```

Microsoft SQL schema definition

The following example shows the SCHEMAS.csv format:

```
name,schema_id,principal_id
dbo,1,1
guest,2,2
INFORMATION_SCHEMA,3,3
```

Microsoft SQL synonym definition

The following example shows the SYNONYMS.csv format:

```
name,schema_id,base_object_name
dotSynonym, 21575115,[EXAMPLE].[dbo].[Table.dot]
bracketSynonym, 37575172,[EXAMPLE].[dbo].[Table[1]
qual,69575286,[EXAMPLE].[dbo].[Table]].[s1]
```

Microsoft SQL table definition

The following example shows the ALL_OBJECTS.csv format:

```
name,object_Id,schema_id,type
all_columns,-103,4,V
all_views,-102,4,V
all_objects,-101,4,V
```

Microsoft SQL column definitions

The following example shows the ALL_COLUMNS.csv format:

```
object_id,name,columnid
-464069102,DOMAIN_NAME,42
-464069102,TABLE_CATALOG,43
-464069102,TABLE_SCHEMA,44
```

Microsoft SQL db.properties

Provide information about vendor and repository type:

```
vendor=MYSQL
repoType=CATALOG
```

Microsoft SQL server link-host mapping

The following example shows db.properties if a server link links the MSSQL01_LINK name with the STG database on the MSSQL01 host:

```
vendor=MSSQL
repoType=CATALOG
connection=MSSQL01_LINK=MSSQL01.STG.DBO
```

To specify more than one link, you can append a numeric ID to the connection parameter:

```
vendor=MSSQL
repoType=CATALOG
connection.1=MSSQL01_LINK=MSSQL01.STG.DBO
connection.2=MSSQL02_LINK=MSSQL02.PROD.DBO
```

Netezza column definition

The following example shows the V_JDBC_COLUMNS1.csv file form Netezza _V_JDBC_COLUMNS1 table:

```
TABLE_CAT,SCHEMA_NAME,TABLE_NAME,COLUMN_NAME,DATA_TYPE,ORDINAL_POSITION
DBB,ADMIN,A,A2,4,3
DBB,ADMIN,A,A1,4,2
DBB,ADMIN,A,A0,4,1
DBB,ADMIN,B,B2,1,3
DBB,ADMIN,B,B1,1,2
DBB,ADMIN,B,B0,1,1
```

Netezza alias definition

The following example shows the _V_SYNONYM.csv file form Netezza _V_SYNONYM table:

```
SYNONYM_NAME,REFDATABASE,REFSCHEMA,REFOBJNAME
LOCAL_A_SYN,DBB,ADMIN,A
```

Netezza db.properties file

Provide information about vendor and repository type:

```
vendor=NETEZZA
repoType=CATALOG
```

SAS column definition

The following example shows the COLUMNS.csv file from SAS DICTIONARY.COLUMNS table:

```
Varnum,Name,Libname,MemName
3,A2,Lib1,A
2,A1,Lib1,A
1,A0,Lib1,A
3,B2,Lib1,B
2,B1,Lib1,B
1,B0,Lib1,B
```

SAS db.properties file

Provide information about vendor and repository type:

```
vendor=SAS
repoType=CATALOG
```

Custom stored procedures

You can provide a custom definition of a Microsoft SQL Server stored procedure. You extract the stored procedure from the database, and override it.

You can override a procedure in the scope that must be parsed but causes problems in its original version. For example, you can override a procedure that carries valuable lineage information but has complex conditional statements that you can't evaluate.

To override the database stored procedure definition, prepare an SQL script file with a custom procedure definition, according to the syntax of the current database vendors. You must store the file in a subfolder in the mdrepo in the following location: `host/database`. The name of the subfolder is the default schema name for the defined procedures.

During the processing, the MetaDex server scans each file in the schema directory to identify overriding definitions, regardless of the file name or extension. Each file might contain many procedure definitions. The MetaDex server uses the first occurrence of a procedure definition.

You can qualify the name of the overridden procedure with schema or the schema name is derived from the directory name.

Qualified names

For a HOST01/DB01/SCHEMA01/file.txt file with the following content:

```
CREATE PROCEDURE dbo.overridingProcedure
AS
BEGIN
insert into A select * from B
END

CREATE PROCEDURE overridingProcedure
AS
```

```
BEGIN
insert into C select * from D
END
```

Use the following procedure override:

```
HOST01.DB01.dbo.overridingProcedure
```

```
HOST01.DB01.SCHEMA01.overridingProcedure
```

Duplicated definition

When both procedures have the same fully qualified name: `HOST.DB.SCHEMA.overridingProcedure`, the second definition is ignored.

```
CREATE PROCEDURE overridingProcedure
AS
BEGIN
insert into F select * from G
END

CREATE PROCEDURE overridingProcedure
AS
BEGIN
insert into H select * from J
END
```

Managing file definitions

If the MetaDex server doesn't recognize files used as processes or any columns are missing in the files, the Files.csv file can solve the problem.

In case of unknown files, the MetaDex server searches for the file definitions in the `HOST/Files.csv` directory. If it can't find file definitions there, it searches the MetaDex repository. If definitions are not there, you receive an error message informing that no files were found.

If the files are not registered, you receive an error. Run the files again to register them.

The following MetaDex scanners scan unknown files:

- Microsoft SQL Server Integration Services
- Talend Data Integration
- Microsoft Power BI
- JCL
- IBM DataStage

If there are any missing columns, the Files.csv file allows the registry to search for them in the file.

You must use the following directory structure for Files.csv: `HOST/Files.csv`

The following example shows a sample Files.csv file:

Files.csv

```
"FilePath","ColumnName"
"C:/first.txt", "a"
"C:/first.txt", "b"
"C:/first.txt", "c"
"C:/second.txt", "a"
"C:/second.txt", "b"
```

In this example, the first.txt file has three defined columns: a, b, and c. The second.txt file has two defined columns: a and b.

Graph properties

MetaDex server stores currently processed objects in graph objects.

Some modules can set additional properties to graphs. You can find these properties in the description of each module and you can access them in Module Parameters with `%propertyName%` syntax.

You can also use `%?SourceFile%` syntax. The question mark indicates that property can be undefined. In this case, the MetaDex server replaces property with an empty string.

For example, if the Mapping Document filename pattern is defined as Output_%?SourceFile% and SourceFile property is never defined, the result file name is Output_.

CHAPTER 2

The MetaDex tool

This chapter includes the following topics:

- The MetaDex tool overview, 21
- MetaDex server requirements, 22
- Logging in to the MetaDex tool, 22
- Changing the MetaDex user password, 23
- Projects page, 23
- Configuration page, 25
- Repository page, 27
- Administration page, 29
- Help page, 29
- License Information page, 30
- MetaDex icons, 30

The MetaDex tool overview

The MetaDex tool is a web-based tool hosted on a server. Data administrators use the interface to perform configuration and administration tasks for MetaDex scanning process.

The MetaDex tool home page consists of the following main elements:

- The header with menus that open the following pages:
  Projects

  For more information, see "Projects page" on page 23.

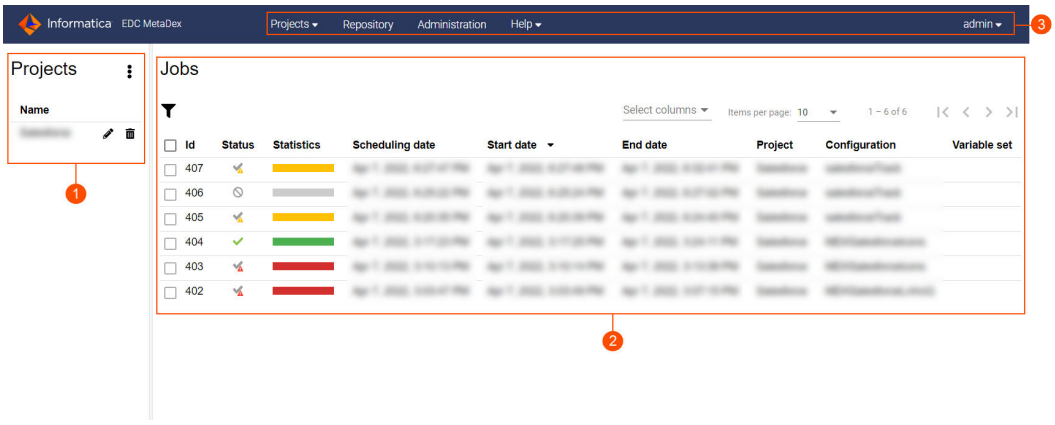
  Repository

  For more information, see "Repository page" on page 27.

  Administration

  For more information, see "Administration page" on page 29.
- The **Projects** area
- The **Jobs** area

The following image shows the details of the MetaDex tool main header:



1. An editable list of current projects.
2. A list of current metadata harvesting jobs together with their status and other details. You can open a job by clicking it. You can filter the results by the id or the status.
3. The menus to access projects, the repository, administration, help, and the administrator profile.

MetaDex server requirements

The following topic lists software requirements for the installation of the MetaDex server.

Software requirements

MetaDex server requires the following resources:

- 1 instance of 4 GB of physical RAM available for the administration processes.
- At least 8 GB of physical RAM available for every parallel instance of a scanner.

  Note: The actual usage depends on the type and volume of the input metadata. If you want to run two parallel conversion processes, you need to provide at least 10 GB of physical RAM, recommended size is 20 GB.
- 1 GB available disk space for installation.
- 20 GB available disk space for temporary files and logs.

Logging in to the MetaDex tool

Use the MetaDex tool URL to log in to the MetaDex tool from a browser.

1. Open a browser.

   For a list of supported browsers, see the applicable Product Availability Matrix (PAM).
2. In the address field, enter the following URL: `http(s)://<MetaDex host>:<port number>`

   where:

   The <MetaDex host> is the server that hosts the MetaDex tool.

8090 is the default port number. The administrator can set a different port number.

3. On the login page, perform the following actions:
   - If you are logging in for the first time, enter the default credentials:
     1. In the **Username** field, enter `admin`
     2. In the **Password** field, enter `admin`
   - Otherwise, enter your *username* and *password*.
4. If you configured authentication with the Informatica domain, select the applicable domain.
5. Click **Log In**.

   The **EDC MetaDex** tool home page appears.

If you are logging in for the first time, to maintain security, change the default password.

Changing the MetaDex user password

Change the MetaDex user password for security reasons.

1. In the top-right corner of the MetaDex tool page, click **admin** > **Change Password**.
2. In the **Change password** window, enter your credentials:
   a. In the **Old password** field, enter `admin`
   b. In the **New password** field, enter your new password.
   c. In the **Confirm new password** field, re-enter your new password.
3. Click **Submit**.

Projects page

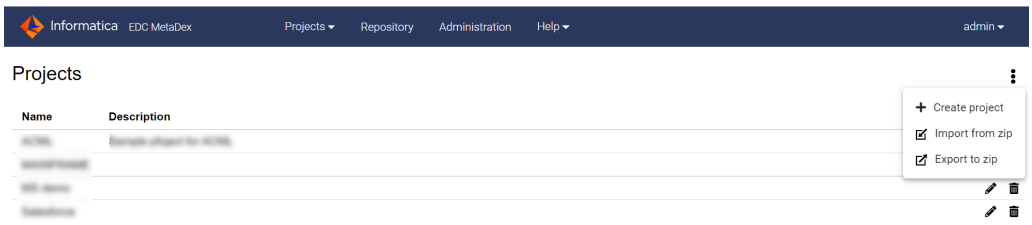
Use the **Projects** page to manage your projects and process metadata.

When you open the MetaDex tool, the **Projects** page shows all your current projects. A project is named after the metadata source it extracts from, such as Oracle. A project is a set of configurations.

By clicking **Projects** in the header, you can display a single project or a list of all of your projects in the MetaDex tool.

On the **Projects** page you can perform the following actions:

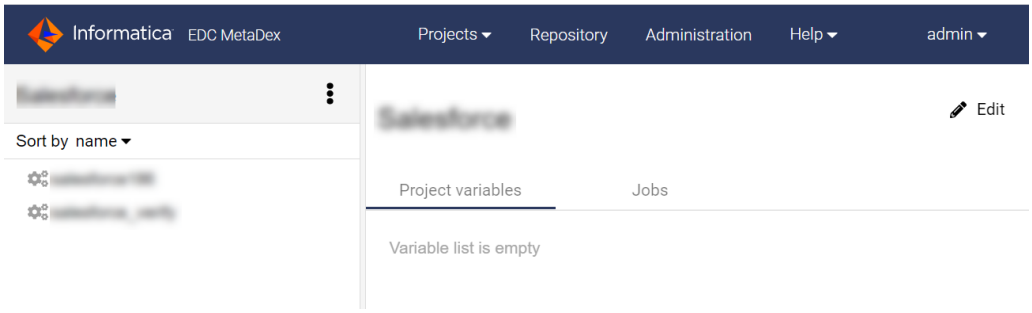
- Selecting **Show all** directs you to a list of current projects.
  The following image shows an example list of current projects in the MetaDex tool:



You can perform the following actions on the projects:

- Add
- Edit
- Remove
- Import from a .zip file
- Export to a .zip file

- Selecting a single project directs you to the project details page.
  The following image shows an example of a project details page in the MetaDex tool:



On the **Project variables** tab, you can view the list of the variables used in the configuration.

Note: By default, the MetaDex tool sanitizes properties that end with security-related words, for example:

- Password
- Pass
- Secret
- Key
- Token
- Credentials

If any sensitive variables are not encrypted, you can add them to the sanitized list by setting the following property: `scanners.security.additional-keys-to-sanitize=<sensitive_variable>`

For more information, see " Supported Extra Properties" in the *MetaDex Scanner Reference*.

On the **Jobs** tab, you can view a list of current metadata harvesting jobs with their status and other details. You can open a job by clicking it. You can filter the results by the ID or the status.

In the top right corner of the window, you can perform the following actions for a project:

- Edit
- Create a configuration
- Import a configuration
- Import a configuration from a .zip file
- Export a configuration to a .zip file
- Configure notifications for a configuration
- Delete a configuration

In the pane on the left, you can see the list of configurations assigned to the project. Clicking on a selected configuration directs you to the configuration page.

Configuration page

Use the **Configuration** page to create and edit the configurations of your jobs. A configuration is a set of parameters that define a metadata scanning job.

You can import and export an existing configuration in a form of a properties file.

You can perform the following actions on the configurations:

- Create a configuration
- Import a configuration
- Import a configuration from a ZIP file
- Export a configuration to a ZIP file

You can perform the following actions for a single configuration:

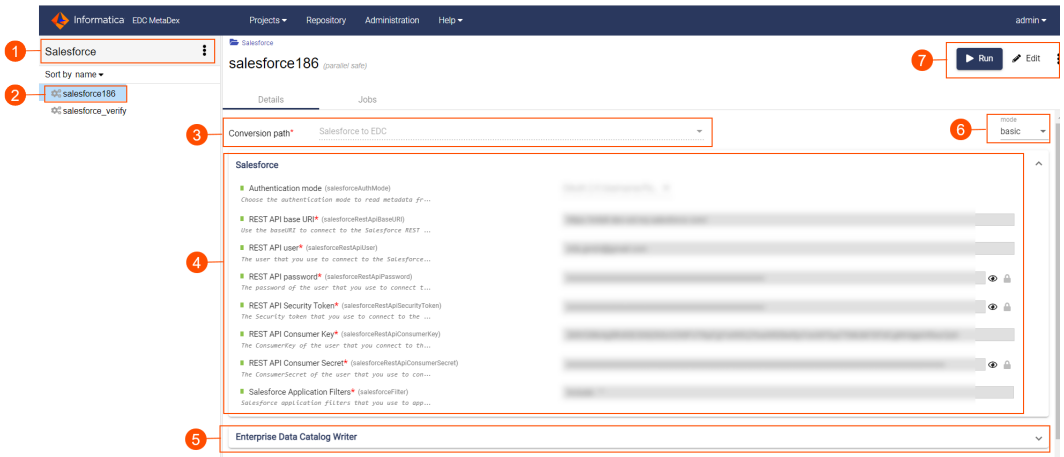
- View details
- View jobs
- Run
- Edit
- Additionally, when you right-click the configuration, you can perform the following actions:
  - Clone
  - Override
  - Import overriding configuration
  - Create a variable set
  - Import a variable set
  - Export
  - Configure notifications
  - Delete

  The following table describes the additional actions you can perform on a configuration:

Parameter	Description
Clone	When you clone a configuration, you create its copy that you can modify. The cloned configuration is at the same level in the hierarchy as the original. Any changes applied to the clone do not affect the original configuration.
Override	When you override a configuration, the MetaDex server creates a child configuration that inherits all the settings from the parent configuration. You can override any settings you want to change in the child configuration and that does not affect the parent configuration, but any changes in the parent configuration also apply to the child configuration.
Import overriding configuration	You import the overriding configuration from a file.

Parameter	Description
Create a variable set	A variable set allows you to run a configuration with different sets of parameters multiple times. A variable set consists of multiple entries. Each entry specifies values for all variables you use in a configuration. When you create a variable set, you determine variables for a configuration.
Import a variable set	You import a variable set from a file.
Export	You export a configuration component to a file.
Configure notifications	You can configure email notifications for a chosen configuration. You can choose if you want to receive notifications after a configuration run results with an error or after every run.
Delete	You delete a configuration component.

The following image shows an example of an expanded configuration details module for the Salesforce MetaDex scanner:



1. The current project and the **More** menu.
2. The list of configurations.
3. The conversion path for the current configuration.
4. An expanded module with parameter fields.
5. Remaining modules.
6. The list of available modes.
7. The buttons to run and edit a configuration, and a **More** menu.

For security reasons, configure the commands that you want to run in a configuration file. The configuration file is located at: ${INFA_HOME}/AdvScannersWorkspace/etc/external-tools-allow-list.properties

You can overwrite default application configurations with custom ones. You can update the MetaDex tool without losing your custom data stored in the workspace, such as the license, external tools you configured or the keystore for the server key. For more information, see "File management" on page 70.

For information about creating and running a configuration, see the *MetaDex Scanner Configuration Guide*.

Parameters

Each configuration has a predetermined set of parameters that is specific to the metadata scanning source it extracts from. You can view and edit parameters. You can define parameters such as which objects to process, where to save outputs, and how to name the resources. When you run a job, it uses the parameters

in the configuration. You can find parameters applicable to your metadata scanning source in the *MetaDex Scanner Reference*.

The default parameters depend on the scanner that you use. For information about modules and their respective parameters, see "Modules Reference" in the *MetaDex Scanner Reference MetaDex Scanner Reference*.

Repository page

On the **Repository** page, you can view job details that are not accessible in the Enterprise Data Catalog.

The MetaDex repository is a database used for processing relevant information, storing assets, and logs produced during scanning.

On the **Repository** page, you can perform the following actions:

- View executables and assets.
  Executables

  The objects that you execute to process data. Sample executables include:

  - Jobs
  - Scripts
  - Programs
  - Processes
  - Reports

  Assets

  The objects that store data. Sample assets include:

  - Files
  - Tables
  - Columns
- Use the search field to find executables and assets containing specific words, for example, customer.

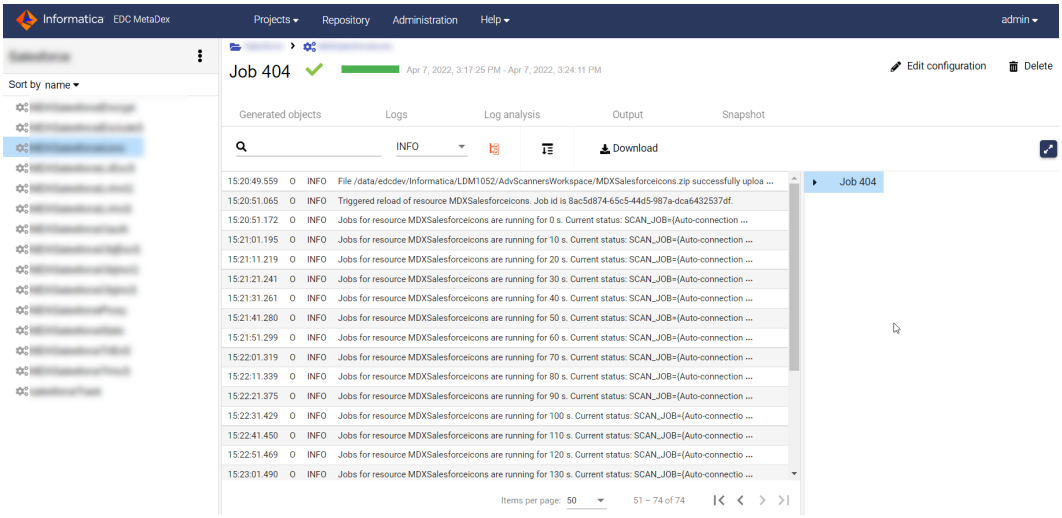
Jobs

A job converts metadata from the source system, based on the parameters in the configuration, and loads it to the destination catalog. You can run jobs from within the MetaDex tool or through external tools configurations.

In the MetaDex tool, you can open a job to view:

- Log details
- Output
- Snapshot showing the parameters that you used
- Objects generated by the job

The following image shows a sample job in the MetaDex tool:



You cannot perform the following operations on MetaDex scanners in the Enterprise Data Catalog tool (the Catalog tool):

- Clone a job
- Schedule a job run
- Run a job

For Snowflake, you can configure the profiling information in the Catalog tool.

Job statuses

The following table lists job statuses and their descriptions:

Status	Description
	The job succeeded with warnings. Review the log files for warning messages. **Note:** Despite the warnings, the scan can still be successful. This status does not mean there are errors in the scan.
	The job was aborted by a user or the system.
	The job succeeded.
	The job completed with errors. Review the log files for error messages.
	The job is waiting to run.
	The job is activating.

Status	Description
	The job is in progress.
	The job failed.

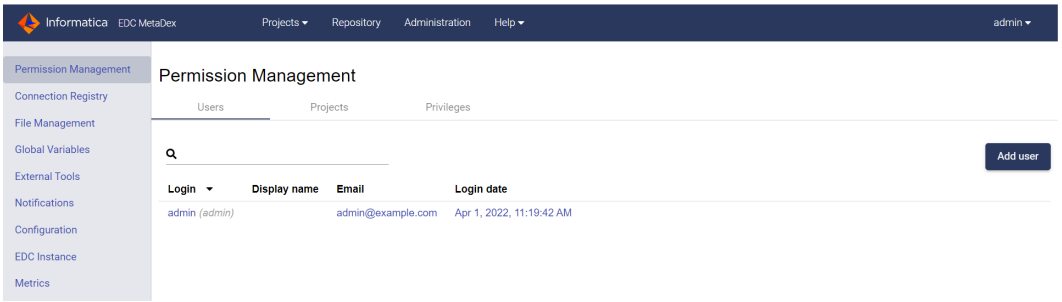
Administration page

Perform the configuration tasks on the **Administration** page.

In the left pane of the **Administration** page, you can select options to manage:

- User permissions
- The Connection Registry
- Files
- Global variables
- External tools
- Notifications
- Configuration
- EDC Instance
- Metrics

The following image shows an example component of the **Administration** page:



Help page

Use the **Help** page to view the information about the MetaDex tool.

Access the **Help** page from the MetaDex tool header. The **Help** page contains the server version information.

License Information page

Use the **License Information** page to view the information about the licensed modules and the license expiration date.

Access the **License Information** page from the MetaDex tool header by clicking **Help** > **License information**.

MetaDex icons

The MetaDex tool uses shortcut icons that help you navigate the interface.

The following table lists the most common MetaDex icons:

Icon	Description
	More. Provides more MetaDex tool options.
	Edit. Opens an editing mode for a given entry.
	Delete. Deletes a given entry, project or job.
	Show filters. Shows filters that you can apply to the entries.
	Next page. Moves to the next page.
	Previous page. Moves to the previous page.
	Last page. Moves to the last page.
	First page. Moves to the first page.

CHAPTER 3

Integration with Enterprise Data Catalog

This chapter includes the following topics:

- Setting the catalog user privilege for the MetaDex tool, 31
- Configuring the MetaDex server, 32
- Enterprise Data Catalog connection parameters, 32
- MetaDex job monitoring, 33
- Configuring custom attributes, 33
- Purging or deleting a MetaDex job, 34

Setting the catalog user privilege for the MetaDex tool

You need a REST API privilege to use the MetaDex tool.

1. Log in to the Informatica Administrator tool.
2. Go to the **Security** tab.
3. From the **Users** list, select the applicable user name.
   The **Overview** tab appears.
4. Select the **Privileges** tab.
5. On the **Privileges** tab, select **Edit**.
   The **Edit Roles and Privileges** window appears.
6. Select **Privileges**.
7. Expand **CS - Catalog Service**.
8. Select **Rest API Privilege** and click **OK**.

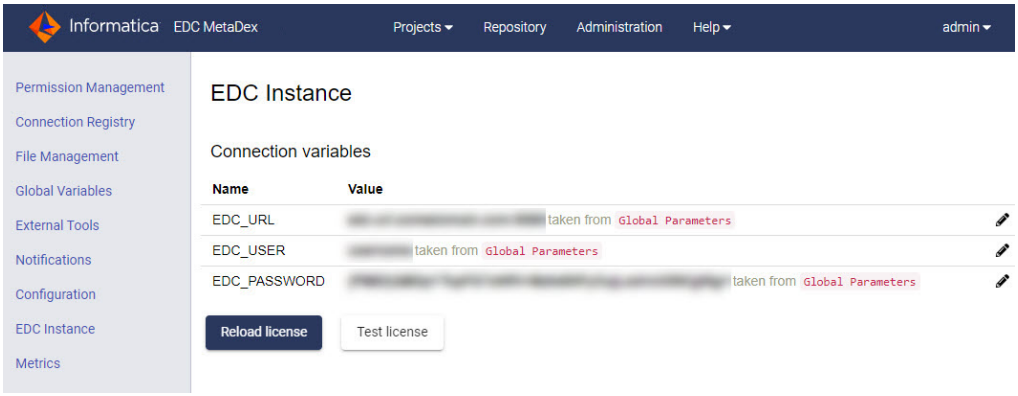
Configuring the MetaDex server

You configure the predefined variables to establish a connection between the MetaDex server and the catalog through the Catalog Service.

For more information about the Catalog Service, see "Catalog Service" in the *Informatica Administrator for Enterprise Data Catalog*.

1. On the header of the MetaDex tool, click **Administration** > **EDC Instance**.

   The **EDC Instance** page appears with the predefined connection variables and the information on the variable origin, as shown in the following image:



2. Click the **Edit** icon to the right of each variable.

   The **Edit global variable** dialog box appears.

3. Enter the values for the following variables:

Name	Description
EDC_URL	The host name and port number of the Catalog Service in the following format: <host>:<port>
EDC_USER	The user name configured for the catalog administrator user.
EDC_PASSWORD	The password configured for the catalog administrator user. When you add the password, click the **Hide password** icon.

4. Click **Save**.

   The **Edit global variable** dialog box closes.

5. Test and reload the license.

Enterprise Data Catalog connection parameters

All MetaDex scanners have an assigned conversion path to Enterprise Data Catalog. Each conversion path has a predetermined set of modules that include parameters that you can configure.

In MetaDex scanners, data sources have a conversion path that you select in the MetaDex tool to load the data into the catalog. The Enterprise Data Catalog Writer module includes connection parameters that allow

you to use in-app visualizations, auto-assign connections, enable reference resources, and keep unresolved reference objects.

For more information about the connection properties for Enterprise Data Catalog, see "Enterprise Data Catalog Writer" in the *MetaDex Scanner Reference*.

MetaDex job monitoring

After a MetaDex job starts sending metadata to the catalog, you can monitor the statistics on the **Monitoring** tab of the Catalog Administrator. You can view a consolidated summary of resources with different task statuses.

The monitoring statistics include the following information:

- Resource schedule
- Task type
- Task status

You can use filters on the **Monitoring** tab to view specific statistics. For more information about monitoring statistics, see the *Informatica Catalog Administrator Guide*.

Configuring custom attributes

You can use the attributes to find relevant information among a large number of data assets and data relationships. MetaDex scanners can have attributes that represent specific properties of object types, such as: Alias Column, Category, and Data Set, based on the data types of the attributes.

1. From the Catalog Administrator tool, click **Library**.

   The **Library** workspace opens.
2. In the resource list, right-click a MetaDex resource in the list and and select **Edit**.

   The MetaDex resource details appear on a new tab.
3. Navigate to the **Custom Attributes** tab within the resource, and configure the attribute settings.
4. Optionally, on the **General** tab, assign custom attribute values to the resource.

   You can choose from the following attributes:

   - Department
   - Data owner
   - Data steward
   - Subject matter expert
5. Click **Save**.

Purging or deleting a MetaDex job

You can purge or delete a MetaDex job from Catalog Administrator.

1. From the Catalog Administrator header, click **Library**.

   The Library workspace opens.
2. In the resource list, hover over the MetaDex resource and, from the control menu, select one of the following options:
   - If you want the resource to disappear from the resource list, select **Delete**.
   - If you want to delete the content of the resource and leave it empty on the resource list, select **Purge**.

CHAPTER 4

MetaDex server administration

This chapter includes the following topics:

- MetaDex server administration overview, 35
- MetaDex server configuration, 36
- Email notifications, 37
- Static content serving, 37
- Setting user roles, 38
- Customizing job settings, 38
- Configure security parameters, 39
- SSL configuration, 39
- Starting and stopping the MetaDex server, 40
- Troubleshooting CBC mode ciphers, 40

MetaDex server administration overview

The MetaDex server interacts with the web-based MetaDex tool.

With the MetaDex server, you can configure and run MetaDex scanners, and work with the logs to identify and fix any problems that arise during the scanning process.

You can, for example, perform the following actions with the MetaDex server:

- Configure email notifications
- Configure static content serving
- Set user roles
- Customize job schedules
- Configure the security parameters

MetaDex server configuration

Use scanners.properties and server-setup.env files to configure the server.

server-setup.env

Use this file to configure environment variables for the server.The server-setup.env file is the source of configuration properties. For example, you can set the value of the SCANNERS_SERVER_JAVA_OPTS variable to configure the server Java settings.

Server-setup.env files are read hierarchically. If you overwrite Java settings in this file, the Java settings from the unpacked application are not valid anymore.

By default, the server-setup.env file is available in the following location: ${INFA_HOME}/services/CatalogService/AdvancedScannersApplication/app. You can also create it in the ${INFA_HOME}/AdvScannersWorkspace directory to store user settings.

scanners.properties

Note: Use the scanners.properties file to configure the server.

The scanners.properties file is available in the following location:

```
<Informatica installation directory>/AdvScannersWorkspace/etc/scanners.properties
```

Both the server and the scanner read this configuration. You can use this file to perform actions such as the ones on the following list:

- "Email notifications" on page 37
- "Static content serving" on page 37
- "Setting user roles" on page 38
- "Customizing job settings" on page 38
- "Configure security parameters" on page 39

Note: In ${INFA_HOME}/AdvScannersWorkspace, server-setup.env file is not available by default. You can create it to store user settings.

While running a scanner job, first load the properties from the configuration script from ${INFA_HOME}/services/CatalogService/AdvancedScannersApplication/app, then from ${INFA_HOME}/AdvScannersWorkspace. This order provides flexibility to define basic environment variables in SCANNERS_HOME\server-setup.env and keep the remaining configuration in the ${INFA_HOME}/AdvScannersWorkspace/server-setup.env file.

The default MetaDex server memory is 4 GB.

${INFA_HOME}/services/CatalogService/AdvancedScannersApplication/app and ${INFA_HOME}/AdvScannersWorkspace are environment constants. ${INFA_HOME}/services/CatalogService/AdvancedScannersApplication/app is a path to the MetaDex tool, and ${INFA_HOME}/AdvScannersWorkspace is a path to the MetaDex server settings.

The MetaDex server consumes properties in a listed order. If both files contain the same environment value, the ${INFA_HOME}/AdvScannersWorkspace file takes precedence. If you want the ${INFA_HOME}/services/CatalogService/AdvancedScannersApplication/app file to take precedence, use its value in the environment definition.

When you upgrade to a later version, user settings in ${INFA_HOME}/AdvScannersWorkspace are preserved.

Email notifications

Enable email notifications by specifying mail sender address in the scanners.properties file.

You can configure notifications globally for all configurations in a project or one configuration. The email addresses of the recipients can contain variables that you resolve basing on the environment, project variables, and variable set.

Configure the following Simple Mail Transfer Protocol server properties:

```
spring.mail.host=smtp.example.com
spring.mail.port=587
spring.mail.username=login
spring.mail.password=password
spring.mail.properties.mail.smtp.auth=true
spring.mail.properties.mail.smtp.starttls.enable=true
```

Mail sender address configuration

```
scanners.web.mailSenderAddress=user@example.com
```

Static content serving

Static content represents files hosted by the web server and exposed to users through URLs. Before in-app visualisations were available, static content was the standard approach to make static HTML files created during scans to be made available for end users. Currently, static content serving is rarely used, but is maintained for backward compatibility. For example, you can share a PDF with usage rules for MetaDex WebUI using static content serving.

You can serve any filesystem content to the users within the Informatica domain groups defined in the `scanners.properties` file. Static content serving by default requires a repository viewer permission. You can configure the Informatica domain groups with a scanners.web.edc.repositoryViewerGroups property in the scanners.properties file. To define more groups. separate them with a semi-colon. For more information on the `scanners.properties` file, see "MetaDex server configuration" on page 36.

You can also serve folders permitting viewing the content with no authentication required with scanners.server.content.authenticationEnabled=false.

By default, the reports directory from the workspace is served to the users. Another directory can be set with the `scanners.server.content.locations` property.

The following example shows the default property:

```
scanners.server.content.locations=reports=scanners.workspace.path/reports
```

To set multiple locations, separate definitions with a semi-colon:

```
scanners.server.content.locations=reps=scanners.workspace.path/reports;extra=/
data/extra
```

The setting exposes the following endpoints:

```
http(s)://server:8090/content/reps/(relative/path/within/workspace/reports
folder)
http(s)://server:8090/content/extra/(relative/path/within/data/extra folder)
```

To disable serving default directory, set an empty locations value: scanners.server.content.locations=

Setting user roles

Authenticated users can access the MetaDex tool if they have at least one assigned role or are members of groups.

You can also assign user roles in the MetaDex tool. See "Assigning roles " on page 69.

1. Navigate to the following location: <informatica installation directory>/AdvScannersWorkspace/etc/scanners.properties
2. Create your configuration.

   The following example shows a sample configuration set in the scanners.properties file:

   ```
   scanners.platform.auth.web.admin-users=Administrator
   scanners.platform.auth.web.admin-groups=Domain_Name/Group_name
   scanners.platform.auth.web.repository-viewer-groups=Domain_Name/Group_name
   ```

   Note: The default separator is a semi-colon. The setup is not intended for command line access. Define this configuration in the scanners.properties file on the MetaDex tool.

Customizing job settings

You can change the default settings of metadata harvesting jobs by adjusting their parameters.

To customize job settings, modify the scanners.properties file with the following parameters:

1. To customize the default number of concurrently running tasks, in the scanners.properties file, set the value for the scanners.server.execution.slots parameter.

   For more information on the `scanners.properties` file, see "MetaDex server configuration" on page 36.

   All execution requests are queued as the number of jobs that can be run at the same moment is limited. The default number of concurrently executing tasks is 2. The minimum is 1. The maximum is usually between 10 and 20, depending on the size of the jobs and the performance of your computer.

   For example, scanners.server.execution.slots=3.
2. If you want to customize the time after which the job status changes from `starting` to `running`, in the scanners.properties file, set the value for the scanners.server.processings.startupTimeout parameter. When the host machine performance is slow, consider increasing the timeout for scanners.server.processings.startupTimeout.
3. If you want to customize the time after which the job is aborted, in the scanners.properties file, set the value for the scanners.server.processings.abortTimeout parameter.

Configure security parameters

To ensure that you can use the MetaDex server securely, you can customize the security parameters in the scanners.properties file.

The following table describes the scanners security parameters:

Parameter	Description
scanners.web.security.login.failedLoginsLimit=5	Defines the number of allowed failed login attempts.
scanners.web.security.login.failedLoginsAllowedIntervalInMin=5	Defines the allowed time between failed login attempts in minutes.
scanners.web.security.login.accountFreezeTimeoutInMin=15	Defines the number of minutes that the account is blocked for after the configured number of failed login attempts.

For more information on the `scanners.properties` file location, see "MetaDex server configuration" on page 36.

SSL configuration

SSL enablement results with the MetaDex tool no longer supporting plain HTTP connections.

You can configure SSL by setting `server.ssl.` properties in the `${INFA_HOME}/AdvScannersWorkspace/etc/scanners.properties` file.

File example:

```
server.port=8443
server.ssl.key-store=keystore.p12
server.ssl.key-store-password={PBE0}C2Kw8KtA21+LbRPgOBNDrN11IhBp+dqkJr493jTRAoI=
server.ssl.key-store-type=PKCS12
server.ssl.key-alias=scanners
```

Location of the keystore is a path to the file. In the example, it is a relative path pointing to `${INFA_HOME}/services/CatalogService/AdvancedScannersApplication/app/keystore.p12`.

To perform a security test, generate a keystore with keytool utility tool available in most operating system:

```
keytool -genkey -alias scanners -storetype PKCS12 -keyalg RSA -keysize 2048 -
keystore keystore.p12
```

You need a store with signed certificate for production purposes.

The default SSL configuration of the MetaDex tool supports cipher block chaining (CBC) mode ciphers. If you receive an error message during security testing, see "Troubleshooting CBC mode ciphers" on page 40.

Starting and stopping the MetaDex server

Use the server.sh command to start and stop the MetaDex server. You can run the command as a root or non-root user.

1. If you run the command as a root user, ensure that you set the SCANNERS _USER variable.

   For information, see "Setup environment" on page 73.

   Note: The start script terminates if you don't set the SCANNERS_USER variable for the root user or if the user specified with that variable doesn't exist. The username specified in the SCANNERS_USER variable must match the operating system username that is executing the process.

2. Open `${INFA_HOME}/services/CatalogService/AdvancedScannersApplication/app`

   a. To start the server, enter `sh server.sh start` or `./server.sh start`

      Note: If you run the command as a non-root user, the MetaDex server starts with nohup as the current user.

   b. To stop the server, enter `sh server.sh stop` or `./server.sh stop`

Troubleshooting CBC mode ciphers

When I perform security testing, the following error message appears:

`TLS/SSL Server Supports The Use of Static Key Ciphers (SSL-static-key-ciphers)`

CBC ciphers are not considered secure by certain standards. The default SSL configuration of the MetaDex tool supports cipher block chaining (CBC) mode ciphers.

To resolve the error, in the `${INFA_HOME}/AdvScannersWorkspace/etc/scanners.properties` file, allow only the following strong cipher modes for SSL:

```
server.ssl.ciphers=TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384,
TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256,
TLS_DHE_RSA_WITH_AES_256_GCM_SHA384,
TLS_DHE_RSA_WITH_AES_128_GCM_SHA256
```

CHAPTER 5

Setting up connectivity

This chapter includes the following topics:

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- Creating a connection, 42
- Types of connection scans, 43
- JDBC connections, 44
- Connectivity to Oracle databases, 49
- Connectivity to DB2 databases, 51
- Connectivity to Google BigQuery databases, 52
- Connectivity to Teradata databases, 53
- Connectivity to Microsoft SQL databases, 54
- Connectivity to Netezza databases, 55
- Connectivity to Sybase databases, 55
- Connectivity to Hive databases, 55
- Connectivity to Snowflake databases, 56
- Connectivity to Amazon Redshift databases, 57
- Connectivity to Greenplum databases, 57
- Connectivity to Databricks databases, 58
- Connectivity to other databases, 59

Connection requirements

To extract metadata from metadata sources to Enterprise Data Catalog, configure drivers and connections.

In the following sample scenarios, you configure connections to access the metadata sources:

- To view the lineage for an RDBMS table, fetch the definition of the table from the metadata source.
- To view the lineage that includes a stored procedure run on an asset, fetch the stored procedure definition from the metadata source.

Creating a connection

A connection consists of a host and a database within that host. To create a connection, add hosts and database entries to the connection registry.

You can use a host to group multiple database instances on the same database server.

1. In the MetaDex tool, on the header, click **Administration**.
2. On the left pane, click **Connection Registry**.

   The **Connection Registry Entries** list is displayed in the right pane.
3. Click **Add host**.

   The **New host** window appears.
4. Enter the new host details:

Property	Description
Host	Required. Enter the host name.
Property	Select the required property. The **Value** field changes into a list.
Value	Select the required value.

5. Click **Save**.

   The **Host** window appears.
6. Click **Add entry**.

   The **New database** window appears.
7. Enter the following database details:

Property	Description
Database	Required. Enter the database name.
Repository type	Required. Select the repository type from the list. **Note:** Use only **CATALOG** and **JDBC**. Other types are applicable only for testing.
Vendor	Required. Select the required vendor.
URL	Required. Enter the JDBC URL. For more information, see "JDBC examples" on page 44.
User	Required only for the JDBC type. Enter the JDBC user name.
Password	Required only for the JDBC type. Enter the JDBC password.
Property	Select the required property.
Value	Select or enter the required value.

8. Click **Save**.

9. You can add more databases by performing steps 6 through 8.

   For more information about the use of properties such as edc.filePathToResourceName and edc.fileConnectionMode in Enterprise Data Catalog, see Update the connection registry.

Types of connection scans

If you don't have connection information to run a connection-aware scan, you can run a connectionless scan.

If you don't enter connection details in the **Data source connections** parameter fields in a scanner configuration, a scan generates lineage based on predictions of the database metadata. Lineage information based on predictions might be incomplete.

If you enter connection details, scans connect to the database and generate complete lineage based on the database metadata.

For example, you want to scan Microsoft PowerBI metadata. The MetaDex job connects to 200 databases, and you have the connection details of only 5 of them. When you run the job, the scan extracts metadata for these 5 databases with complete lineage. The job runs a connectionless scan on the other 195 databases.

Connectionless scans

Connectionless scans can result in incomplete or inaccurate lineage based on predictions of the repository metadata.

Without the connection information, the MetaDex job processes as much of the lineage source as possible. A warning message about a connectionless scan appears in the log, as well as other warnings about incorrect or incomplete lineage. The MetaDex job then sends all information it receives to the Catalog tool.

Run connectionless scans to identify configuration issues before you run a full scan or if you don't require a full scan.

Connectionless scans have the following disadvantages:

- Might generate inaccurate lineage with fewer details.
- Provide less coverage.
- Take longer to complete than connection-aware scans.
- Don't support PL/SQL calls or metatables that require access to the database.

Connection-aware scans

Informatica encourages you to set up connections to your databases so that you can run connection-aware scans to retrieve complete lineage information.

The configuration must include the following the data source connection values:

- Database connection key
- Resolved MDREPO entry
- Defined connection registry entry

The MetaDex job sends all information it receives to the Catalog tool.

JDBC connections

JDBC is the Java API that manages connecting to a database, issuing queries, commands, and handling result sets obtained from the database.

With db.properties files, you can specify the JDBC connections to fetch the catalog metadata automatically and directly from the original database. These files describe connection parameters for a single database and must be in the database level of the registry.

You can encrypt passwords with encrypt.bat or encrypt.sh encryption program. No additional configuration options are required for storing encrypted passwords. The application automatically recognizes and internally decrypts the passwords. For more information, see "Password encryption" on page 65.

JDBC examples

Use the parameter descriptions and examples from this section to configure the JDBC connection.

Database definition parameters

The following table describes the database definition parameters:

Parameter	Description
url	Required. JDBC connection string.
user	Required. The name of the user.
password	Required. In plain text or encrypted with encrypt.sh or encrypt.bat utility.
driverClass	Fully qualified JDBC driver class, for example, org.netezza.Driver.

connection	Required if the target database connects to other databases. Provides mappings between the database links of the target and other MetaDex connection registry entries.
	You can use the following syntax to specify the connection:
	`connection.ID=CONNECTION_KEY=[VENDOR]:`
	`[USER@]HOST.DATABASE.[SCHEMA]`
	For example:
	`connection=MSSQL01_LINK=MSSQL-HOST.STG.DBO`
	To specify more than one link, you can append a numeric ID to the connection parameter:
	`connection.1=MSSQL01_LINK=MSSQL-HOST.STG.DBO`
	`connection.2=(HOST=abc.com)=ORACLE:ORA-H.ORA-DB.ORA-S`
	`connection.3=REGEXP:.*abc.*=ORACLE:ORA-H.ORA-DB.ORA_S`

repository.fetchStrategy	Fetches objects from the database. You can set it to one of the following values: - Default is ALL_OWNER_OBJECTS. Fetches all database objects, such as tables, views, and procedures in a particular schema to build a cache and optimize performance at the cost of increased memory consumption. - SINGLE_OBJECT. Fetches the required objects slower but with lower memory usage. Supported by Oracle and Teradata vendors.

Parameter	Description
initialSql	Specifies the SQL query executed on each connection to the database. You can use it in conjunction with initialSqlFile. The query is run every time the MetaDex server connects to the database. The query does not return any results and does not cause any side effects when executed multiple times.
initialSqlFile	Specifies the path to a file that contains SQL statements separated by a semicolon, executed when MetaDex server connects to the database. The file can be either relative to the db.properties file or absolute. You can use it in conjunction with initialSql. The query does not return any results and does not cause any side effects when executed multiple times.
catalogSchema	Overrides the default catalog tables schema.
catalogTables	Overrides the default catalog table.
vendorVersion	Stores the database version. Use only for IBM Db2 for z/OS.
db2zos.useDB2DatabaseNames	Set to true indicates that the MetaDex server uses information stored in the DBNAME column in the SYSTABLES catalog table to determine database name for a particular table. The default behavior is to use information expressed by the connection registry directory structure.

Sample JDBC connections

Examples of basic JDBC configuration:

Oracle

```
url = jdbc:oracle:thin:@oracle01.exampleorg.local:1521:cspl
user = telephone
password = telephone
vendor=ORACLE
repoType=JDBC
```

Oracle with an overridden driverClass

```
url = jdbc:oracle:thin:@oracle01.exampleorg.local:1521:cspl
user=orauser
password=orapass
vendor=ORACLE
repoType=JDBC
driverClass=oracle.jdbc.OracleDriver
```

Oracle with a single-object fetch strategy

```
url = jdbc:oracle:thin:@oracle01.exampleorg.local:1521:cspl
user=orauser
password=orapass
vendor=ORACLE
repoType=JDBC
repository.fetchStrategy=SINGLE_OBJECT
```

Oracle with specified post-connect queries

```
user=orauser
password=orapass
vendor=ORACLE
repoType=JDBC
initialSql=alter session set "_common_data_view_enabled"=TRUE;
initialSqlFile=SetupEnv.sql
```

IBM Db2

```
url = jdbc:db2://127.0.0.1:50000/SAMPLE
user = telephone
password = telephone
vendor=DB2
repoType=JDBC
```

IBM Db2 z/OS

When you extract the view definitions from a Db2 to a z/OS database, you must specify the version. Valid values are: `8`, `9`, `10` and `11`.

```
url = jdbc:db2://127.0.0.1:50000/SAMPLE
user = telephone
password = telephone
vendor=DB2ZOS
vendorVersion=10
repoType=JDBC
db2zos.useDB2DatabaseNames = true
```

Teradata

```
url = jdbc:teradata://127.0.0.1/1025/Teradata01
user = telephone
password = telephone
vendor=TERADATA
repoType=JDBC
```

Microsoft SQL Server

```
url = jdbc:sqlserver://localhost:1433;databaseName=<DatabaseName>
user = telephone
password = telephone
vendor=MSSQL
repoType=JDBC
```

Netezza

```
url = jdbc:netezza://192.168.153.128:5480/DBA
user = telephone
password = telephone
vendor=NETEZZA
repoType=JDBC
netezzaSchemasEnabled=false
```

Apache Hive

```
url=jdbc:hive2://host:port/database
repoType=JDBC
vendor=HIVE
user = user
password = pass
```

Snowflake

```
repoType=JDBC
vendor=SNOWFLAKE
url=jdbc:snowflake://account.snowflakecomputing.com?warehouse=<name>&db=<name>
user=user
password=password
```

Amazon Redshift

```
repoType=JDBC
vendor=REDSHIFT
url=jdbc:redshift://example.com:5439/database
user=user
password=password
```

Greenplum

```
repoType=JDBC
vendor=GREENPLUM
url=jdbc:postgresql://example.com:15432/database
user=user
password=password
```

SAP Hana

```
repoType=JDBC
vendor=HANA
url=jdbc:sap://example.com
user=user
password=passwordd
```

Microsoft Azure Data Warehouse (Synapse)

```
repoType=JDBC
vendor=SYNAPSE
url=jdbc:sqlserver://account.database.windows.net:1433;database=<name>;
user=user
password=password
```

Testing the JDBC connections

You can verify the JDBC connections in the MetaDex repository.

1. Open the directory where you installed the MetaDex tool.

2. From the command line, run the `connregtest` script for your operating system"

connregtest.bat

The following example shows a sample script:

```
>connregtest.bat
Usage:
connregtest.bat <mdrepopath> <option>+
Possible options:
* connregtest.bat <mdrepopath> -list (initializes repository and lists
all host.db combinations with their repo types, db types,
jdbc URLs and usernames)
* connregtest.bat <mdrepopath> -test host.db (finds host.db in mdrepo and
verifies it)
* connregtest.bat <mdrepopath> -test ALL (finds all db.properties in
mdrepo and verifies them)

>connregtest.bat d:\AdvancedScanners\mdrepo -list
...
------------------------------
HOST.DB    REPOTYPE     VENDOR     USER     URL
HH.BB      JDBC         TERADATA   u1       jdbc:teradata://terahost
H2.B2      JDBC         ORACLE     u2
jdbc:oracle:oci:@oraclehost:1521:sid

>connregtest.bat d:\AdvancedScanners\mdrepo -test HH.BB
...
HH.BB       : SUCCESS
```

connregtest.sh

The following example shows a sample script:

```
connregtest.sh /home/AdvancedScanners/mdrepo -list
```

JDBC drivers

MetaDex provides most of the required JDBC drivers in the installation package.

The installation package contains licensed DataDirect JDBC drivers for the following databases:

- Oracle
- IBM Db2
- Microsoft SQL Server
- Sybase ASE
- MySQL
- PostgreSQL

DataDirect JDBC drivers are ready to use. You don't need to download or install any additional components.

Note: MetaDex provides Microsoft JDBC Driver for SQL Server which is not a DataDirect JDBC driver. Informatica recommends that you use it instead of the DataDirect driver.

If you apply Informatica 10.5.3.0.1 , the following changes take effect when you configure the Microsoft JDBC Driver for SQL Server scanner:

- You must use the following user name and password properties instead of the aadSecurePrincipalId and aadSecurePrincipalSecret connection properties:

  - scanners.repository.username=<value of AADSecurePrincipalId>

  - scanners.repository.password=<value of AADSecurePrincipalSecret>

  Note: If the password is not encrypted, you must use the encrypted value of the principal secret connection property.
- To resolve a specific realm, you can use the CNAME resolution for the realm.
- Microsoft JDBC Driver for SQL Server uses `encrypt=true` by default. You can disable encryption if the database connection is not encrypted. Use `encrypt=false` in the JDBC URL to disable encryption.

  Note: Informatica recommends that you always encrypt the connection to the database.
- If you use `encrypt=false` in the JDBC URL, and the Microsoft SQL Server database mandates encryption, you must validate the certificate using the trustServerCertificate connection setting.

If needed, download other drivers such as Netezza or Teradata from the vendor website. You can then create a jdbc folder and copy the JDBC driver files into the following location: `$SCANNERS_WORKSPACE/jdbc` or `$SCANNERS_WORKSPACE/lib/jdbc`

Important: When you connect to the data source, specify the connection string in the following format:

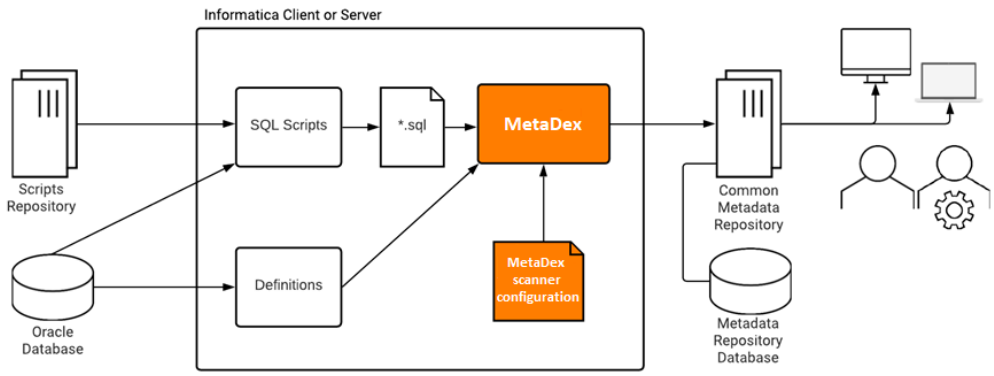
```
jdbc:<subprotocol>:<subname>
```

If you specify "informatica" as a subprotocol, MetaDex selects the DataDirect driver included in the Informatica installation package. For example, if you enter `jdbc:informatica:oracle://<host>:<port>;SID=<value>` MetaDex selects the DataDirect Oracle JDBC driver, but if you enter `jdbc:oracle:thin:@//<host>:<port>:<SID>` MetaDex uses a native Oracle JDBC driver.

Connectivity to Oracle databases

The MetaDex server has particular dependencies to process metadata related to Oracle databases, such as SQL scripts, view definitions, or ETL jobs.

The following image depicts the MetaDex integration with the Oracle database:



- SQL scripts contain PL/SQL logic

- Definitions contain:

  View definitions

  Table definitions

  Synonyms

  DB link definitions

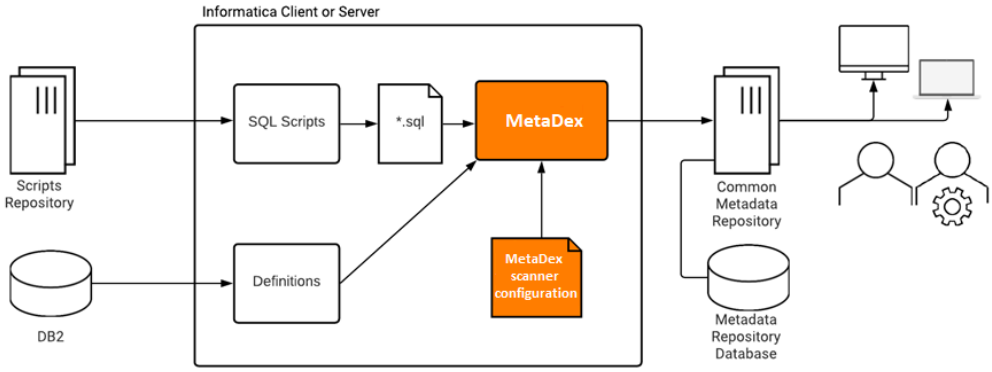
The MetaDex server has the following dependencies:

- For a single non-RAC Oracle Database installation with native authentication, the MetaDex server uses a JDBC type 4 thin driver. The default port is 1521.
- The MetaDex server uses JDBC type 2 thick driver for a RAC and RADIUS-authenticated Oracle Database installation:
  - Oracle Database Client is installed and configured on the host where the MetaDex tool is installed.
  - Network connectivity required for the Oracle Database Client is operational. Detailed connectivity requirements depend on the Oracle Database version, and you can verify them in the Oracle documentation.
- Oracle database credentials with permissions that allow you to perform the following operations:
  - select on SYS.DBA_ARGUMENTS
  - select on SYS.DBA_COL_COMMENTS
  - select on SYS.DBA_COLL_TYPES
  - select on SYS.DBA_CONS_COLUMNS
  - select on SYS.DBA_CONSTRAINTS
  - select on SYS.DBA_DB_LINKS
  - select on SYS.DBA_DIRECTORIES
  - select on SYS.DBA_EXTERNAL_LOCATIONS
  - select on SYS.DBA_EXTERNAL_TABLES
  - select on SYS.DBA_MVIEWS
  - select on SYS.DBA_OBJECTS
  - select on SYS.DBA_OPBINDINGS
  - select on SYS.DBA_SEQUENCES
  - select on SYS.DBA_SOURCE
  - select on SYS.DBA_SYNONYMS
  - select on SYS.DBA_TAB_COLUMNS
  - select on SYS.DBA_TAB_COMMENTS
  - select on SYS.DBA_TABLES
  - select on SYS.DBA_TRIGGERS
  - select on SYS.DBA_TYPE_ATTRS
  - select on SYS.DBA_USERS
  - select on SYS.DBA_VIEWS
  - select on SYS.NLS_SESSION_PARAMETERS
- Grant a predefined SELECT_CATALOG_ROLE role

Connectivity to DB2 databases

The MetaDex server has particular dependencies to process metadata related to DB2 databases, such as SQL scripts, view definitions, or ETL jobs.

The following image depicts the MetaDex integration with the DB2 database:



- SQL scripts contain SQL logic
- Definitions contain:

  View definitions

  Table definitions

The MetaDex server requires the following dependencies:

- Network connectivity between any port on the machine where you installed the MetaDex tool and the listening port of the DB2 Database, the default port is 50000.
- By default, the MetaDex server uses a JDBC type 4 driver to connect to the DB2 Database. If required, you can reconfigure the MetaDex server to use a JDBC type 2 thick driver.

  Note: When you connect the driver to the DB2 database for the first time, ensure that you have the following additional privileges:

  •BINDADD

  •DBA for NULLID

  The privileges allow the driver to create all required DB2 packages when it connects to the database. You can revoke the additional privileges after the first connection.
- You must install the DB2 Database client software on the machine where you installed the MetaDex server.
- Perform the following operations with UDB DB2 database credentials with permissions:

  - select on syscat.COLUMNS

  - select on syscat.FUNCTIONS

  - select on syscat.NICKNAMES

  - select on syscat.PROCEDURES

  - select on syscat.PROCPARMS

  - select on syscat.TABLES

  - select on syscat.USEROPTIONS

- select on syscat.VIEWS
- select on syscat.SCHEMATA
- select on syscat.TABCONST
- select on syscat.REFERENCES
- select on syscat.KEYCOLUSE

- Perform the following operations with DB2 for z/OS database credentials with permissions:
  - select on sysibm.SYSCOLUMNS
  - select on sysibm.SYSPARMS
  - select on sysibm.SYSROUTINES
  - select on sysibm.SYSSEQUENCES
  - select on sysibm.SYSSYNONYMS
  - select on sysibm.SYSTABLES
  - select on sysibm.SYSVIEWS
- Perform the following operations with ISeries DB2 database credentials with permissions:
  - select on sysibm.SQLCOLUMNS
  - select on sysibm.SQLTABLES

Connectivity to Google BigQuery databases

The MetaDex server has particular dependencies to process metadata related to Google BigQuery databases.

Note: Effective in version 10.5.5, Google BigQuery scanner is available for technical preview.

Technical preview functionality is supported for evaluation purposes but is unwarranted and is not supported in production environments or any environment that you plan to push to production. Informatica intends to include the preview functionality in an upcoming release for production use, but might choose not to in accordance with changing market or technical circumstances. For more information, contact Informatica Global Customer Support.

The MetaDex server has the following dependencies:

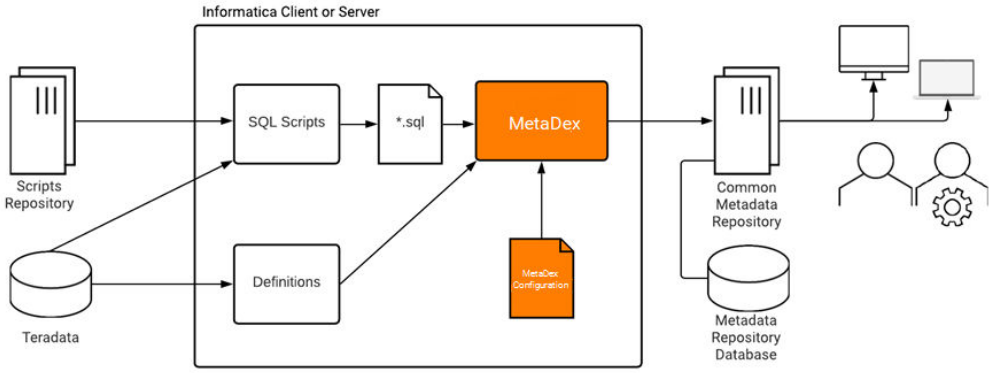
- Network connectivity between any port on the machine where you installed the MetaDex tool and the listening port of the Google BigQuery database. The default port is 443.
- To extract metadata from a Google BigQuery source system, you can choose from one of the following options:
  - If you want to use the existing roles and avoid creation of custom roles, assign the BigQuery Data Viewer and BigQuery Metadata Viewer roles to the service account that you use to access the Google Cloud Platform project.
  - If you want to use minimal permissions, create a custom role with the following permissions and assign the custom role to the service account that you use to access the Google Cloud Platform project:
    - bigquery.datasets.get
    - bigquery.models.getMetadata

- bigquery.routines.get
- bigquery.tables.get
- bigquery.tables.list
- bigquery.jobs.create
- bigquery.tables.getData

Connectivity to Teradata databases

The MetaDex server has particular dependencies to process metadata related to Teradata databases, such as SQL scripts, stored procedure or view definitions, or ETL jobs with embedded queries.

The following image depicts the MetaDex integration with DB2database:



- Definitions contain:

  View definitions

  Table definitions

The MetaDex server requires the following dependencies:

- Network connectivity between any port on the machine where you installed the MetaDex tool and the listening port of the Teradata Database (default port: 1025).
- Teradata database credentials with permissions allowing to perform the following operations:
  - select on dbc.COLUMNSV
  - select on dbc.TABLESV
  - select on dbc.TABLETEXTV
  - select on dbc.INDICESV
  - select on dbc.ALL_RI_CHILDRENV
- Teradata database non-table credentials with permissions that allow you to perform the following operations:
  - SHOW PROCEDURES. Permission to access definitions of database stored procedures. Required if processing stored procedures is in the project scope.

- SHOW MACROS. Permission to access definitions of database macros. Required if processing macros is in the project scope.
- HELP PROCEDURES. Permission to get information about the parameters and their attributes. Required if processing stored procedures is in the project scope.
- HELP MACROS. Permission to get information about the parameters and their attributes. Required if processing macros is in project scope.

Connectivity to Microsoft SQL databases

The MetaDex server has particular dependencies to process metadata related to Microsoft SQL databases, such as SQL scripts, view definitions, or ETL jobs.

The MetaDex server requires the following dependencies:

- Network connectivity between any port on the machine where you install the MetaDex tool and the listening port of the Microsoft SQL database. Default port is 1433. By default, the MetaDex server uses a provided JDBC type 4 driver to connect to the Microsoft SQL database.
- Microsoft SQL Server database credentials with permissions that allow you to perform the following operations:

  SELECT ON sys.all_columns

  SELECT ON sys.all_objects

  SELECT ON sys.all_parameters

  SELECT ON sys.database_principals

  SELECT ON sys.databases

  SELECT ON sys.foreign_key_columns

  SELECT ON sys.indexes

  SELECT ON sys.index_columns

  SELECT ON sys.partitions

  SELECT ON sys.schemas

  SELECT ON sys.sql_modules

  SELECT ON sys.synonyms

  SELECT ON sys.tables

  SELECT ON sys.types

  SELECT ON sys.table_types, if your database uses table types.

Connectivity to Netezza databases

The MetaDex server has particular dependencies to process metadata related to Netezza databases, such as SQL scripts, view definitions, or ETL jobs.

The MetaDex server has the following dependencies:

- Network connectivity between any port on the machine where you installed the MetaDex tool and the listening port of the Netezza database. The default port is 5480.
- Netezza database credentials with permissions that allow you to perform the following operations:
  - select on _V_OBJECTS
  - select on _V_PROCEDURE
  - select on _V_SCHEMA
  - select on _V_SEQUENCE
  - select on _V_SYNONYM
  - select on _V_VIEW
  - select on _V_JDBC_COLUMNS1

Connectivity to Sybase databases

To process metadata related to Sysbase databases, the MetaDex tool requires particular dependencies.

MetaDex requires the following dependencies:

- Network connectivity between any port on the machine where you install the MetaDex tool and the listening port of the Sybase database. The default port is 5000.
- Sybase database credentials with permissions that allow you to perform the following operations:
  - select on dbo.syscolumns
  - select on dbo.syscomments
  - select on dbo.sysobjects
  - select on dbo.sysservers
  - select on dbo.systypes
  - select on dbo.sysusers

Connectivity to Hive databases

To process metadata related to Hive databases, the MetaDex server has particular dependencies.

The MetaDex server has the following dependencies:

- Network connectivity between any port on the machine where you installed the MetaDex tool and the listening port of the Hive database. The default port is 10000.

- Hive database credentials with permissions that allow you to perform the following operations:
  - select on sys.COLUMNS_V2
  - select on sys.DBS
  - select on sys.PARTITION_KEYS
  - select on sys.SDS
  - select on sys.TBLS

Connectivity to Snowflake databases

To process metadata related to Snowflake databases, the MetaDex server has particular dependencies.

The MetaDex server has the following dependencies:

- Network connectivity between any port on the machine where you installed the MetaDex tool and the listening port of the Snowflake database. The default port is 443.
- Snowflake database credentials with permissions that allow you to perform the following operations:
  - select on information_schema.EXTERNAL_TABLES
  - select on information_schema.FUNCTIONS
  - select on information_schema.PIPES
  - select on information_schema.PROCEDURES
  - select on information_schema.SCHEMATA
  - select on information_schema.SEQUENCES
  - select on information_schema.STAGES
  - select on SNOWFLAKE.ACCOUNT_USAGE.TAGS
  - select on SNOWFLAKE.ACCOUNT_USAGE.TAG_REFERENCES
  - show objects
  - show columns
  - show primary keys
  - show imported keys
  - show streams
  - show materialized views
  - show tasks
  - show databases

Connectivity to Amazon Redshift databases

To process metadata related to Amazon Redshift databases, the MetaDex server has particular dependencies.

The MetaDex server has the following dependencies:

- Network connectivity between any port on the machine where you installed the MetaDex tool and the listening port of the Amazon Redshift database. The default port is 5439.
- Amazon Redshift database credentials with permissions that allow you to perform the following operations:
  - select on pg_catalog.PG_ATTRIBUTE
  - select on pg_catalog.PG_CLASS
  - select on pg_catalog.PG_CONSTRAINT
  - select on pg_catalog.PG_DESCRIPTION
  - select on pg_catalog.PG_LANGUAGE
  - select on pg_catalog.PG_NAMESPACE
  - select on pg_catalog.PG_PROC
  - select on pg_catalog.PG_TYPE
  - select on pg_catalog.PG_VIEWS
  - select on pg_catalog.SVV_EXTERNAL_TABLES
  - select on pg_catalog.SVV_EXTERNAL_COLUMNS
  - select on information_schema.COLUMNS
  - select on information_schema.TABLES
  - select on pg_catalog.PG_TABLES
  - select on pg_catalog.PG_CLASS_INFO
  - select on pg_catalog.PG_PROC_INFO
- Permissions to SHOW EXTERNAL TABLE on the tables that you want to process.

Connectivity to Greenplum databases

To process metadata related to Greenplum databases, the MetaDex server has particular dependencies.

The MetaDex server has the following dependencies:

- Network connectivity between any port on the machine where you installed the MetaDex tool and the listening port of the Greenplum database. The default port is 5432.
- Greenplum database credentials with permissions that allow you to perform the following operations:
  - select on pg_catalog.PG_ATTRIBUTE
  - select on pg_catalog.PG_CLASS
  - select on pg_catalog.PG_CONSTRAINT
  - select on pg_catalog.PG_DESCRIPTION
  - select on pg_catalog.PG_LANGUAGE

- select on pg_catalog.PG_NAMESPACE
- select on pg_catalog.PG_PROC
- select on pg_catalog.PG_TYPE
- select on pg_catalog.PG_VIEWS
- select on information_schema.COLUMNS
- select on information_schema.TABLES
- select on pg_catalog.PG_TABLES
- select on pg_catalog.PG_MATVIEWS

Connectivity to Databricks databases

To process metadata related to a Databricks database, the MetaDex server has particular dependencies.

The MetaDex server has the following dependencies:

- Network connectivity between any port on the machine where you installed the MetaDex tool and the listening port of the Databricks database. The default port is 443.

  For more information about configuring Databricks JDBC drivers, see Databricks documentation.
- Databricks database credentials with permissions that allow you to perform the following operations:
  - show tables
  - describe table extended
  - show schemas
  - describe schema extended
- To extract Databricks Unity Catalog lineage with system tables, grant access permissions to the following system tables:
  - system.access.column_lineage
  - system.access.table_lineage

  Note: If you don't have access to system tables, the metadata extraction job uses REST APIs to extract lineage data.
- To extract Databricks Unity Catalog table metadata with system tables, set the following permissions:
  - Grant USE CATALOG permission on the Databricks Unity Catalog.
  - SELECT and USE SCHEMA permissions on the INFORMATION_SCHEMA schema of the user-based catalog where you extract database assets.

  The metadata extraction job uses the following INFORMATION_SCHEMA schema tables to extract metadata:
  - TABLES
  - COLUMNS
  - SCHEMATA
  - VIEWS
  - KEY_COLUMN_USAGE
  - REFERENTIAL_CONSTRAINTS

- CATALOGS

Note: If you don't have access to the INFORMATION_SCHEMA tables, the metadata extraction job uses the `Show` and `Describe` commands to extract metadata.

Connectivity to other databases

You can connect to other databases if you perform an installation of the appropriate JDBC and select read-only access to the database catalog.

The exact list of tables and resources defining the database catalog varies according to each database vendor and how the JDBC drivers interact with them.

The MetaDex server can use JDBC drivers of 1, 2, 3, and 4 type with the following prerequisites:

- Installation and configuration of any native components required for type 1, 2, or 3 JDBC drivers
- The proper network connectivity and credentials required by the driver

Note: You must establish a connection between the server on which you installed the MetaDex tool and the database. The connection requires the JDBC driver and the user access data. Depending on how your administrator configured the database, it can use IP-filtering.

CHAPTER 6

MetaDex repository administration

This chapter includes the following topics:

- Repository overview, 60
- Repository properties, 61
- Requirements for repository vendors, 62
- Password encryption, 65
- Viewing asset details with visualization , 65

Repository overview

Use the MetaDex repository database for storing the functional elements of the MetaDex tool.

The MetaDex repository stores the following functional modules:

- Administration
- Projects and jobs
- Repository

You perform the configuration tasks in the **Administration** page and create projects and process metadata in the **Projects and Jobs** areas.

The MetaDex server loads processed metadata into the catalog. If you cannot view all of the job details in the catalog, you can access them on the **Repository** page in the MetaDex tool.

To configure the repository database, use the repository.properties file and the property names listed in "Repository properties" on page 61.

Repository properties

When you configure the MetaDex repository, use the repository.properties file located in the ${INFA_HOME}/AdvScannersWorkspace/etc/ directory.

Database configuration

A database initializes automatically during the MetaDex server startup, and manual schema management is not required.
The following example shows a sample database configuration:

```
scanners.repository.jdbcUrl=jdbc:oracle:thin:@oraclehost:1521:sid
scanners.repository.username=scanners
scanners.repository.password={PBE0}C2Kw8KtA21+LbRPgOBNDrN11IhBp+dqkJr493jTRAoI=
scanners.repository.schema=scanners
```

Properties description

The following table lists all properties that you can use to configure the MetaDex repository:

Note: You do not have to change the default values.

Property name	Description
scanners.repository.jdbcUrl	A JDBC URL. Specifies the location of the database, which the MetaDex server uses for storing processing data. For more information, see "JDBC connections" on page 44.
scanners.repository.schema	The schema used by the MetaDex repository. Default value depends on the underlying data management system. - For Oracle, the default schema is the username specified in the scanners.repository.username property. - For Microsoft SQL Server, the default schema is dbo.
scanners.repository.username	The user name for the database is specified by scanners.repository.jdbcUrl. Default is empty.
scanners.repository.password	The user password for the database specified by scanners.repository.jdbcUrl. The password can be plain or encrypted with the encrypt.sh encryption program. For more information, see "Password encryption" on page 65. Default is empty.
scanners.repository.storeSerializedExecutables	Use this property to save serialized graph definitions to enable visualizations. Default is true.
scanners.repository.batch.insert.buffer.size	Maximum size of the buffer. When the buffer is filled, it splits into batches and is saved into the database. Default is 10000.
scanners.repository.batch.insert.size	The size of a single batch saved in the database. For example, if buffer_size is set to 5000 and batch to 100, there are 50 batches saved one by one in the database. Default is 100.
scanners.repository.batch.select.size	Limits the size of a single set used with SQL IN operator. Default is 25.

Property name	Description
scanners.repository.[|log.][server|job].minIdleConnections	The minimum number of idle connections to maintain in the pool. The default for the server is set to three for the main repository and two for logs. For jobs, three for the main repository and one for the log.
scanners.repository.[|log.][server|job].maxConnections	The maximum number of concurrent connections to the database. For the server, the default is set to 10 for the main and 3 for the log repository. For jobs, three for main and two for the log.
scanners.repository.[|log.][server|job].idleConnectionTimeout	Specifies the time a connection can be in the idle pool. Is retried if the number of idle connections is bigger than the specified minimum idle connections number. Applies if server.minIdleConnections is less than server.maxConnections.
scanners.repository.[|log.][server|job].connectionMaxTimeout	Controls the maximum lifetime of a connection in the pool. Configure to be shorter than any database connection time limit. Default is 30 minutes.
scanners.respository.askBeforeUpdate	Determines whether to prompt before updates of the internal repository if there are new repository changes. Default is false. False. Does not prompt before updates to the database. True. Prompts for confirmation before applying updates.

Requirements for repository vendors

The repository stores the resource configuration and data domain information for the MetaDex server.

View the requirements and configuration examples for the following repository database vendors:

- Oracle
- Microsoft SQL Server

Oracle

Database requirements

Ensure that the database meets the following requirements:

- User privilege to create:
  - SESSION
  - TABLE
  - SEQUENCE
  - VIEW

  Important: Do not use periods in the name of a schema. For example, `name.surname`.
- Tablespace quota depending on system or custom use. Recommended size is 10 GB.
- Assigned tablespace in case of custom use.

You can use either a system tablespace or create a custom one:

- The following example configurations shows how to use system tablespace for the *exampleuser* user name:

```
create user exampleuser identified by password;
grant create session to exampleuser;
alter user exampleuser quota 10g on system; -- quota for system tablespace
usage
grant create table to exampleuser;
grant create sequence to exampleuser;
grant create any view to exampleuser;
```

- The following example configurations shows how to use system table space for the *exampletab* custom tablespace:

```
create tablespace exampletab datafile 'exampletab.dat' size 1 autoextend
on;
create temporary tablespace exampletab_temp tempfile 'exampletab_temp.dat'
size 5M autoextend on;
create user exampleuser identified by password default tablespace
exampletab temporary tablespace exampletab_temp;
grant create session to exampleuser;
grant create table to exampleuser;
grant create sequence to exampleuser;
grant create any view to exampleuser;
grant unlimited tablespace to exampleuser;
```

Configuration example

Sample repository.properties file for Oracle:

```
scanners.repository.jdbcUrl=jdbc:informatica:oracle://oraclehost:1521;SID=sid
scanners.repository.username=scanners
scanners.repository.password={PBE0}C2Kw8KtA21+LbRPgOBNDrN11IhBp
+dqkJr493jTRAoI=
scanners.repository.schema=scanners
```

Note: In Oracle databases, schemas and users are interconnected. You must either use the same user and schema name or configure the schema.

Microsoft SQL Server

Database requirements

Ensure that the database meets the following requirements:

- A user is created in the used database.
- Assigned privilege for the schema is used in the database.

  Important: Do not use periods in the name of a schema.
- User privilege for creating:

  - TABLE

  - VIEW

  - REFERENCES
- User privilege for ALTER command.

Sample configuration for *exampleuser*.

```
GRANT CREATE TABLE TO exampleuser;
GRANT CREATE VIEW TO exampleuser;
GRANT ALTER TO exampleuser;
CREATE SCHEMA exampleschema AUTHORIZATION exampleuser;
```

Microsoft SQL Server by default blocks the table from reading when other transactions modify the table. To ensure that parallel jobs work, you must change the database settings to allow reading from the table that is modified. Change the database settings by running the following command:

```
ALTER DATABASE <DATABASE_NAME>
SET READ_COMMITTED_SNAPSHOT ON WITH ROLLBACK IMMEDIATE
```

Note: The MetaDex repository cannot be a case-sensitive SQL Server database.

Configuration examples

Sample repository.properties file for Microsoft SQL Server.

```
scanners.repository.jdbcUrl=jdbc:sqlserver://docker01:51433;databaseName=mdx
scanners.repository.username=sa
scanners.repository.password=<password>
scanners.repository.schema=dbo
```

Note: The password must meet the criteria for encrypted passwords.

Sample repository.properties file for Microsoft SQL Server with the domain login.

```
scanners.repository.jdbcUrl=jdbc:sqlserver://
docker01:51433;integratedSecurity=true;authenticationScheme=NTLM;domain=NAMEOF
DOMAIN;databaseName=mdx
scanners.repository.username=sa
scanners.repository.password=<password>
scanners.repository.schema=dbo
```

Note: The password must meet the criteria for encrypted passwords.

Sample repository.properties file for Microsoft SQL Server with Azure Active Directory login.

```
scanners.repository.jdbcUrl=jdbc:sqlserver://
azuredb:51433;authentication=ActiveDirectoryPassword;encrypt=false;trustServer
Certificate=true;domain=NAMEOFDOMAIN;databaseName=mdx
scanners.repository.username=sa
scanners.repository.password=<password>
scanners.repository.schema=dbo
```

Note: The password must meet the criteria for encrypted passwords.

If Windows Authentication is required from a Linux machine, use the JDBC driver provided by Microsoft for SQL Server. You cannot use the jTDS driver to connect to the SQL server instance when you use it as a repository database. The driver is obsolete.

Password encryption

All passwords you provide during scanners configuration must be encrypted.

Use the encryption mechanism in the MetaDex tool wherever the **Encrypt** button is enabled. The encryption you use is "`PBEwithSHA256and128bitAES`."

When you work with textual files, encrypt passwords with encrypt.sh, located in the `ScannersHome` directory.

Viewing asset details with visualization

Use the visualization option to analyze the details of assets loaded by the MetaDex server to the catalog.

You can also view asset details with visualization in the MetaDex tool, on the **Generated Objects** tab. See "Viewing job details" on page 74.

1. Log in to Enterprise Data Catalog.

   See "Logging in to Enterprise Data Catalog" in the *Enterprise Data Catalog User Guide*.
2. Find your asset.

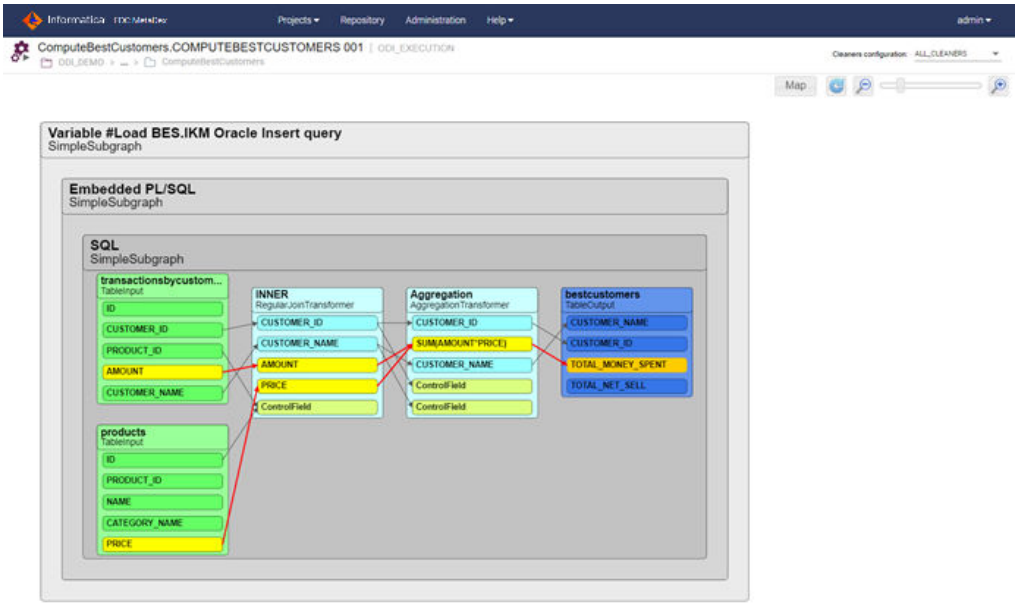
   See "Searching for Assets" in the *Enterprise Data Catalog User Guide*.
3. Click the asset to view the details.
4. On the asset details page, in the **System Attributes** section, click the **Visualization** link.

   For more information about System Attributes, see "View Asset Details Overview" in the *Enterprise Data Catalog User Guide*.
5. Log in to the MetaDex tool.

   The MetaDex tool displays the detailed visualization of the Enterprise Data Catalog asset. The asset opens from the MetaDex repository database in a form of a graph.

6. In the graph, double-click each element to display the detailed data flow of your asset.

   With the expanded visualization view, you can understand relationships between the data in your asset. For example, the value of TOTAL_MONEY_SPENT is the result of multiplying the AMOUNT value by the PRICE value, as shown in the following image:



CHAPTER 7

Job administration

This chapter includes the following topics:

- Job administration overview, 67
- Using the MetaDex tool, 67
- Setup environment, 73
- Viewing job details, 74
- Viewing job log messages, 75
- Filtering, reviewing and downloading job logs, 76
- Removing jobs, 77
- MetaDex logs and reports, 78
- Repository view for job details, 78
- Configuring LDAP authentication , 79
- Data upload, 81
- Content Security Policy, 82
- Enrichment Migration Utility, 82

Job administration overview

This chapter is intended to help a data administrator in an enterprise to run MetaDex jobs.

To run a job, you configure MetaDex scanners in the MetaDex tool.

Using the MetaDex tool

The MetaDex tool is a web-based tool that you use to administer MetaDex scanners.

You can connect to the web server by a configured port. All operations related to data, jobs, and database connections are performed on the machine where the web server is run.

Testing the connection registry entries

You can test the connection registry entries through the MetaDex tool.

1. In the MetaDex tool. on the header, click **Administration**.
2. On the left pane, click **Connection Registry**.

   The **Connection Registry Entries** list is displayed in the right pane.
3. Perform one of the following actions:
   - If you want to test all of the available entries, click **Test entries.**
   - If you want to test a specific entry, click the **check mark** icon.

   The **Connection registry test results** window appears with the test result.

Creating a user

1. In the MetaDex tool, click **Administration**.
2. From the menu, select **Permission Management**.
3. On the **Users** tab, click **Add user**.

   The **New user** window appears.
4. In the **New user** window, enter the following information:
   - Login
   - Password
   - Display Name
   - Email address
5. Click **Save**.

Assigning privileges

Use the MetaDex tool to assign privileges. You can assign Admin or Repository Viewer privilege to users or groups. Unlike roles, they are global permissions assigned to users or groups, not within a project.

1. In the header of the MetaDex tool, click **Administration**.
2. From the menu, select **Permission Management**.
3. On the **Users** or **Groups** tab, from the list, select the domain name.
4. Select a login from the list.

   The configuration page for the login appears.
5. To assign privileges, perform the following actions:
   a. Click **Privileges**.
   b. Use the slider to assign privileges and click **Save**.

The following table lists privileges and the associated permissions:

Privilege	Permissions
Admin	- Can assign roles for users and groups. - Can create, edit and delete projects. - Has operator permissions in each project.
Repository Viewer	- Can access the static content generated by the MetaDex server.

Assigning roles

Use the MetaDex tool to assign roles. Roles provide users with additional permissions to manage projects, configurations, and jobs. You can assign a role to a user within a particular project.

If you configured native authentication through the Informatica domain, you can also assign roles in the scanners.properties file. For more information, see "Setting user roles" on page 38.

1. In the header of the MetaDex tool, click **Administration**.
2. From the menu, select **Permission Management**.
3. On the **Users** tab, from the list, select the domain name.
4. Select a login from the list.

   The configuration page for the login appears.
5. To assign roles, perform the following actions:
   a. Click **Assign roles**.

      The **Edit roles** window appears.
   b. To add a project, click **Add** next to the project name.

      The project appears in the **Roles** section.
   c. Use the slider to assign roles in your project and click **Save**.

The following table lists roles and the associated permissions:

Role Name	Permissions
Viewer	- Can view project content.
Operator	- Has viewer permissions. - Can run configurations and variable sets. - Can configure email notifications. - Can abort and delete jobs.
Developer	- Has operator permissions. - Can create, edit , and delete configurations and variable sets. - Can upload, edit, and delete files located in allowed data locations on the server side.
Owner	- Has developer permissions. - Can edit projects.

File management

A workspace is a dedicated folder for application configuration. On the **File management** tab, you can administer files stored in the workspace.

In a workspace, the server stores the following components:

- Reports
- Temporary files
- Debug logs
- Other application configuration files

For example, you can use debug logs stored in the workspace to inspect errors.

You can also overwrite default application configurations with custom ones. You can easily update the MetaDex tool without losing your custom data stored in the workspace, such as the license, external tools you configured or the keystore for the server key.

By default, you can find the workspace folder in the following location: ${INFA_HOME}/AdvScannersWorkspace

Related Topics:

- "Data upload" on page 81

Resetting user passwords

An administrator can reset a password for internal source users.

1. In the header of the MetaDex tool, click **Administration**.
2. From the menu on the left, select **Permission Management**.
3. From the domain list, select **MetaDex**.
4. On the **Users** tab, select the required **Login**.

   The **Users** window appears.
5. In the **Users** window, click the **More** icon.
6. Click **Reset Password**.

   A **Reset password** prompt appears.
7. From the **Reset password** prompt, copy the reset password URL.
8. Send the reset password URL to the user.

   The link remains active for 24 hours.

To reset the password, the user can click the link and enter the login and a new password.

Synchronizing users and user groups

Use the MetaDex tool to synchronize users and user groups. MetaDex automatically synchronizes users and user groups in Enterprise Data Catalog every two hours.

Additionally, you can manually synchronize users and user groups using the **Synchronize** option on the **Authentication Sync** tab.

Important: You must configure the INFA_HOME variable and the scanners.properties file before you synchronize users and user groups. For more information, see "Configuring the authentication" on page 80.

1. In the header of the MetaDex tool, click **Administration**.
2. From the menu, select **Permission Management**.
3. On the **Authentication Sync** tab, click **Synchronize**.

   The MetaDex tool displays the status of the synchronization.

External tools

You can use external tools with the MetaDex tool to facilitate tasks.

By default, the list of external tools is empty. If you do not create this configuration file, you cannot run any external tool.

If your MetaDex tool has external tools configured, you can use the external tools in the same way as the MetaDex tool.

For security reasons, configure the commands that you want to run in a configuration file. You can find the configuration file in the following location: `${INFA_HOME}/AdvScannersWorkspace/etc/external-tools-allow-list.properties` Create the file if it is not available in this location.

Note: To apply the changes, restart the MetaDex server after you modify the external-tool-allow-list.properties file.

The file contains a list of allowed patterns. You can perform the following actions:

- To expand to any number of characters on the same directory level, use *
- To expand to any number of characters across directory levels, use **

For example, `/opt/scripts/*.sh` allows executing all .sh scripts, excluding subdirectories, while `/opt/scripts/**.sh` allows executing .sh scripts in subdirectories.

You can pass the following arguments to external tools:

- To pass the file paths as an arguments, enter `**`
- If you do not want to pass the file paths as arguments, enter `*`

The following entry enables that you use an external tool that runs scripts with one or more arguments from the `/opt/scripts/` directory:

```
/opt/scripts/sampleScript.sh **
```

To use zero or more arguments, create an entry with and an entry without a single asterisk or double asterisks:

```
/opt/scripts/sampleScript.sh
/opt/scripts/sampleScript.sh **
```

Adding external tools

Use the MetaDex tool to add external tools.

1. In the MetaDex tool, go to **Administration**.
2. Select **External Tools**.
3. Click **Add external tool**.

4. In the **New external tool** section, enter the tool details.

   The following table describes the tool properties:

Property	Description
Name	Required. The name of the tool that you want to add.
Tool ID	Required. The ID of the tool. Default is the name of the tool.
Command	Required. The command that launches the tool.
Description	The description of the tool.
Working directory	The path to the directory.
Log layout	Describes the format of logs: date, severity, and message. You can use it to parse data from standard output, save it properly to the database and work with it in the same way as with MetaDex logs. If the log layout is empty or a log does not match it, everything is treated as a message and the log is saved with the current date and INFO severity. Standard error data is not saved to the database. The following example shows sample layout format: `%d{yyyy-MM-dd HH:mm:ss} %p %m` The following example shows a sample message: `2018-11-02 14:34:02 ERROR` *`error message`*
Icon	Defines displayed image for the tool to differentiate configurations visually. If you do not select it, a default icon is used.
Config Schema	.JSON schema you can use to generate a .JSON editor during external tool configuration. If you do not define it, you can set the configuration content in the text area.

5. Click **Save**.

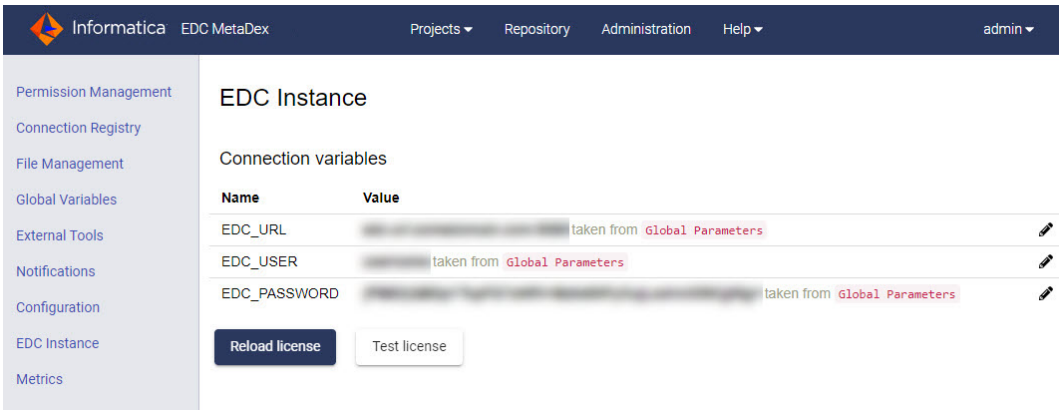
Access to the Catalog Service

On the **Administration** tab, you can access the **EDC Instance** page.

The **EDC Instance** page contains predefined variables required to establish a connection between the MetaDex server and the following Enterprise Data Catalog tools:

- The Catalog tool
- The Catalog Administrator tool

The following image shows the **EDC Instance** page:



Note: If you define any of the connection variables as an environment variable, you can see its value with information about the variable origin. For variables that are not derived from the environment nor global parameters, you can see an empty placeholder for a variable.

To verify the connection to the Enterprise Data Catalog instance, click the **Test license** button.

If the connection is successful, to activate the license, click the **Reload license** button.

Setup environment

Use the setup.env file to define variables for every startup of the MetaDex server. For example, setting the value of the *SCANNERS_JAVA_PROPERTIES* variable to `-Xmx8g` defines the startup memory as 8 GB.

You can store any user settings for the execution environment in setup.env in the following locations:

- ${INFA_HOME}/services/CatalogService/AdvancedScannersApplication/app
- ${INFA_HOME}/AdvScannersWorkspace

In ${INFA_HOME}/AdvScannersWorkspace, setup.env is not available by default. You can create it to store user settings.

When you upgrade to a later release, MetaDex, user settings in ${INFA_HOME}/AdvScannersWorkspace are preserved.

After each execution of the application, first load the properties from the configuration script from ${INFA_HOME}/services/CatalogService/AdvancedScannersApplication/app then from ${INFA_HOME}/AdvScannersWorkspace. Reading the setup.env file in this order provides flexibility to define basic environment variables in ${INFA_HOME}/services/CatalogService/AdvancedScannersApplication/app/setup.env and keep the remaining configuration in the ${INFA_HOME}/AdvScannersWorkspace/setup.env file.

To override the configuration script for a specific MetaDex scanner, you can configure the **Execution options** under **Additional options**. You can override settings, such as memory, Java heap size, and JVM stack size for a specific MetaDex scanner configuration.

Viewing job details

View details of the executables and assets that are scanned while processing metadata. You can expand each object to view its hierarchy in the MetaDex repository.

1. In the MetaDex tool, from the **Jobs** area, open a job by clicking it.

   A job details window appears.

2. Click the **Generated objects** tab.

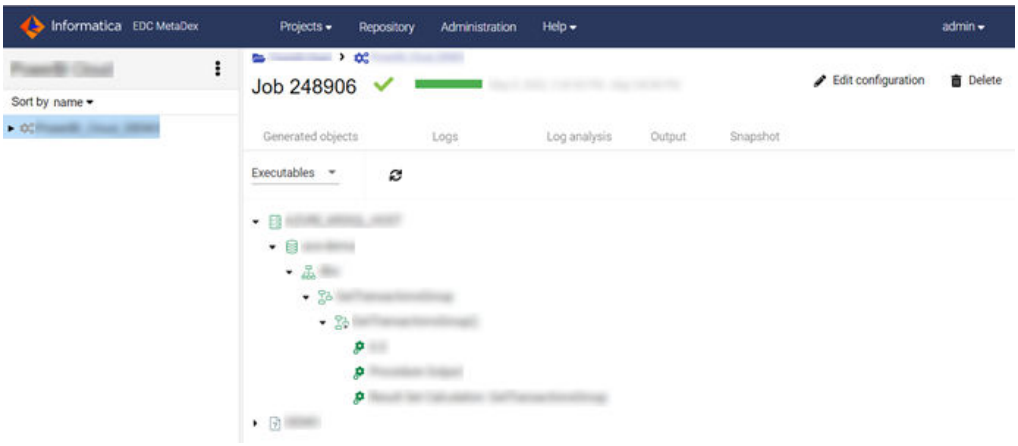
3. From the list, select one of the following types of generated objects:

   - Executables
   - Sources and Targets
   - Assets

     Note: The "Assets" option allows you to see statistics for non-lineage assets encountered during a job.

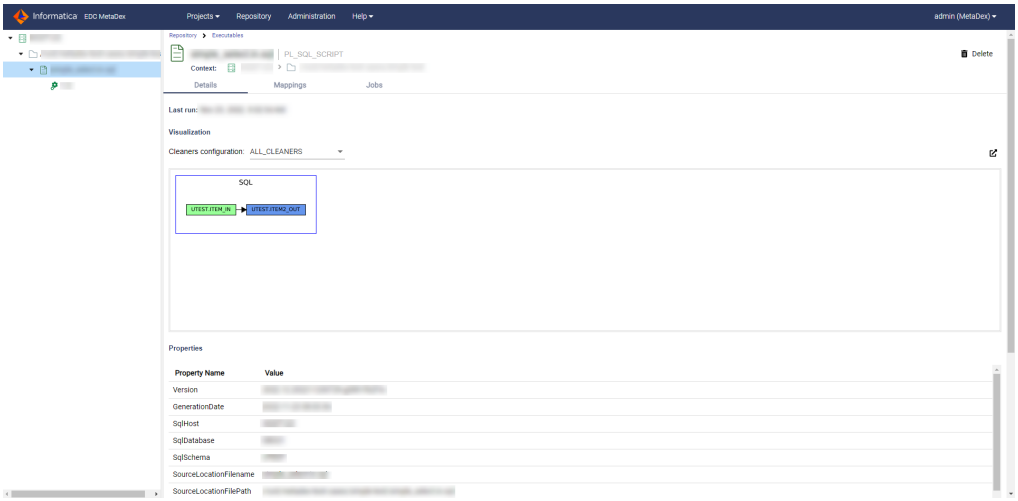
4. Expand an item to see the hierarchy of objects.

   The following image shows a fully expanded generated object:



5. Click an object to open it in the MetaDex repository and see more details.

   The following image shows a statement object in the MetaDex repository with a visualization:



Based on the object type, you can see the following information:

Data processes information	Assets information
Asset visualization	Assets and executables that are impacted by the object
Source and target assets	Assets and executables that are impacted by the object
Children	Children

a. In the left pane, expand the directory tree to see the localization of the object in the MetaDex repository.

b. On the **Jobs** tab, view the jobs that generated the chosen object.

c. If mappings for the object are available, click the **Mappings** tab to view and compare the following elements of jobs:

- Target columns
- Source columns
- Expression

Viewing job log messages

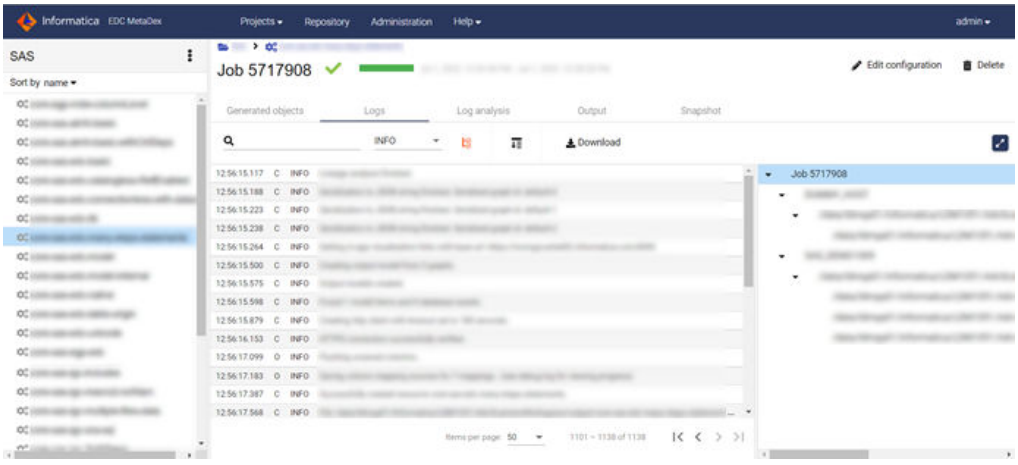
View the log messages that a MetaDex job creates when you run a job. You can expand the structure of a job to view detailed log messages for each object in the job.

1. In the MetaDex tool, from the **Jobs** area, open a job.

   A job details window appears.

2. Click the **Logs** tab.

3. In the right pane, expand the **Job** list, as shown in the following image:



4. Click an object.

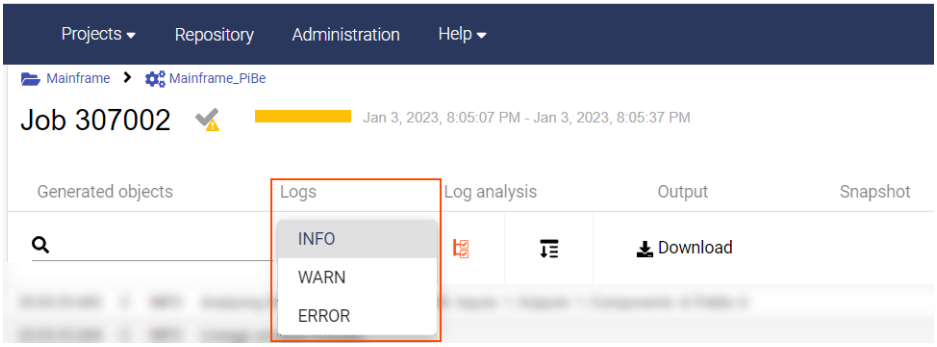
   Log messages for the object appear in the job log details pane.

Filtering, reviewing and downloading job logs

In the MetaDex tool, you can filter log messages, search for logs that contain a given phrase, see statistics for all jobs, analyze the statistics bar for specific jobs, and download debug log files.

Reviewing and filtering log messages

To see job details in the MetaDex tool, select a value from the **Logs** list, as shown in the following image:

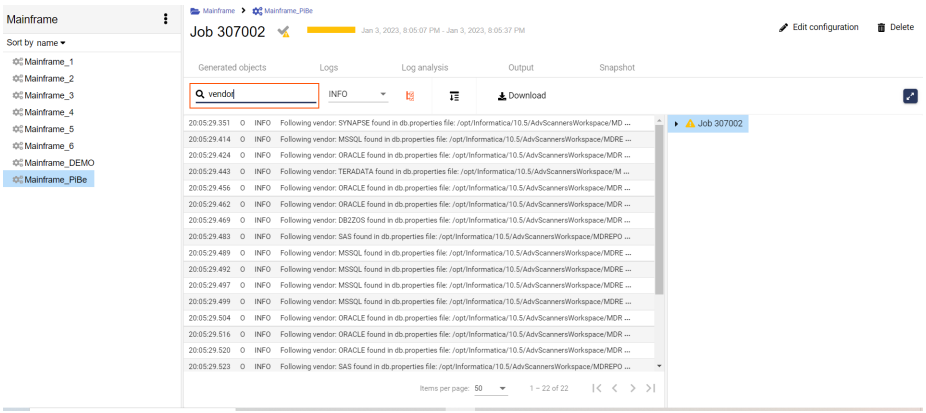


The values in the **Logs** list represent different log levels. The following table describes the log levels:

Value	Description
INFO	Provides additional information about a job. Includes all messages.
WARN	Informs about a warning. The job still succeeds. Includes warnings and errors.
ERROR	Informs about an error that made the job fail. Includes ERROR messages only.

There might be jobs with no messages at a given log level.

Use the search box to filter and review the logs for a particular job, as shown in the following image:



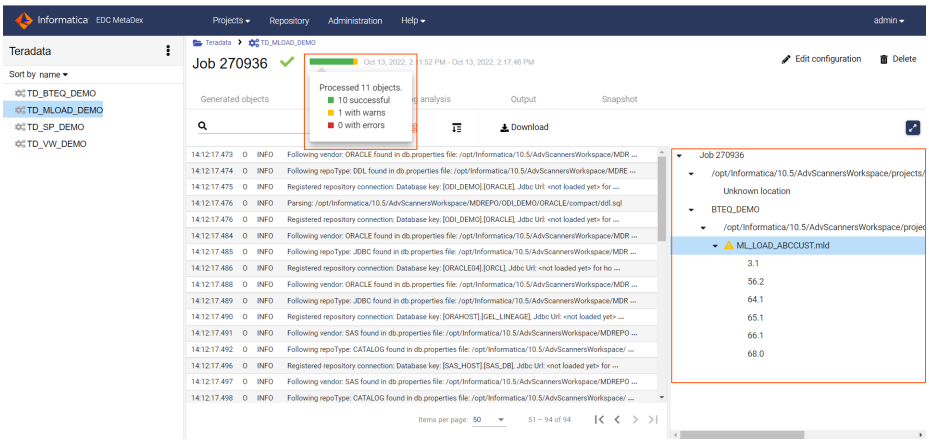
To see the number of processed objects, select a job and refer to the job statistics bar. The objects fall into the following categories:

- Processed successfully.

- Processed with warnings.
- Processed with errors.

INFO, WARN, and ERROR messages attach to generated objects and their children. To see log messages which apply to a given object and its children, click on the object in the panel on the right.

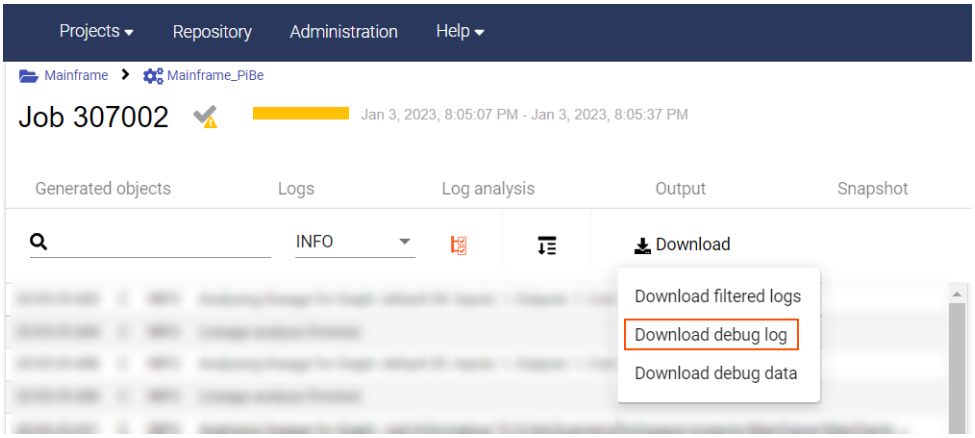
The following image illustrates the statistics bar and the generated objects:



Debug logs

To review lower levels of log information, download a debug log. To download the debug log, select a job and click **Logs** > **Download** > **Download debug log**.

The following image shows the **Download debug log** option:



Removing jobs

To prevent the database from running out of space, enable a removal process that analyzes all finished jobs and removes the oldest.

The removal process occurs in the background. Based on an expression delivered by a job scheduler utility, such as cron, the process preserves the latest jobs defined by the scanners.repository.processings.history.max parameter in the repository.properties file. Default is 3.

Note: By default, the job scheduler is enabled with the following values:

```
#preserve 3 newest processings per blueprint/varset
scanners.repository.processings.history.max=3
#run every 6 hours
scanners.repository.processings.history.cron=0 0 */6 * * *
```

1. Log in to the Linux instance as root.
2. From the command prompt, enter `${INFA_HOME}/AdvScannersWorkspace/etc/repository.properties`
3. In the repository.properties file, set the properties of the scanners.repository.processings.history.cron parameter:
   a. To run the removal process once per hour, enter `scanners.repository.processings.history.cron=0 0 * * * *`
   b. To run the removal process every three hours, enter `scanners.repository.processings.history.cron=0 0 */3 * * *`

MetaDex logs and reports

MetaDex generates logs and conversion reports. Use the logs and conversion reports to debug and troubleshoot issues.

All log files that the MetaDex server generates when MetaDex scanner jobs run are stored in the following directory: `${INFA_HOME}/AdvScannersWorkspace/logs`. You can find daily log files in the `${INFA_HOME}/AdvScannersWorkspace` directory.

If an active MetaDex instance does not have a log file open for writing, you can compress or remove the log files.

When you report any issues with the MetaDex tool to Informatica Global Customer Support, provide the debug log for the job. You can download debug log files from the MetaDex tool. Select the job and click **Logs** > **Download** > **Download debug log**.

Microsoft Excel log reports

You can generate Microsoft Excel reports that contain scanner job logs and objects for MetaDex scanner jobs that you run in the MetaDex tool. To enable reports, set the scanners.reports.xlsx-output property in the scanners.properties file. For more information, see "Supported Extra Properties" in the MetaDex Scanner Reference.

Enabling debug logging level for authentication

1. Navigate to the `WORKSPACE/etc/scanners.properties` configuration file.
2. Enter the following line:
   `logging.level.com.compactsolutionsllc.cdimc.ui.server.security.auth=DEBUG`

Repository view for job details

You can access data describing jobs in the `PUBLIC_PROCESSINGDETAILS` repository view.

The following table describes database columns:

Column Name	Definition
id	Job ID
projectId	Project ID
projectName	The name of the project
configurationId	Configuration ID
configurationName	The name of the configuration
configurationDescription	The description of the configuration
toolId	The ID of the too, for example configuration type
variableSetId	If you created a job by running the variable set, the ID of the variable set
variableSetName	If you created a job by running the variable set, the name of the variable set
variables	The values of variables you use in a job defined for the whole project or in a variable set
status	Job status
startDate	Job start date
schedulingDate	Job scheduling date
endDate	Job finish date
executables	Number of executables
exportedExecutables	The number of executables that were exported
warns	Number of warnings
errors	Number of errors
executor	Login of the user who runs the job

Configuring LDAP authentication

If you want to implement LDAP authentication, you must configure LDAP authentication and synchronize users between the MetaDex framework and the Informatica domain.

Configuring the authentication

Configure the INFA_HOME variable and the scanners.properties file to implement LDAP authentication. The authentication allows you to use the same users and groups in the MetaDex tool and in the Catalog Administator tool.

1. Open the scanners.properties file from the following location: `SCANNERS_WORKSPACE/etc`
2. Add the following credentials for the Catalog Administrator user that has the privilege to manage users and groups:

   ```
   scanners.platform.auth.password=ENCRYPTED_PASSWORD
   scanners.platform.auth.security-domain=SECURITY_DOMAIN
   scanners.platform.auth.username=USERNAME
   ```

3. If you have more than one domain in the domains.infa file, specify the domain that you want to use for user management:

   ```
   scanners.platform.auth.domain=DOMAIN_NAME
   ```

4. If you want to migrate from a previous version of the MetaDex server with LDAP configuration, add the following property:

   ```
   scanners.platform.auth.migration.domain-order-
   search=SecurityDomain1;SecurityDomain2;SecurityDomain3
   ```

   Note: This property specifies the order in which the MetaDex server searches for the LDAP users in the security domains configured on the platform.

5. Run the following commands to recycle the MetaDex server:

   ```
   INFA_HOME/services/CatalogService/AdvancedScannersApplication/app/sh
   server.sh stop
   INFA_HOME/services/CatalogService/AdvancedScannersApplication/app/sh
   server.sh start
   ```

Providing the administrator role to users and user groups

You can use the scanners.properties file to provide the admin role to users and user groups to use them in the MetaDex tool.

Note: You can also assign privileges directly in the MetaDex tool. For more information, see "Assigning privileges" on page 68.

1. Open the scanners.properties file from the following location: `SCANNERS_WORKSPACE/etc`
2. Add the following properties:
   - To set one user as an administrator, add: `scanners.platform.auth.web.admin-users=Native/Administrator`

     Note: Where <Native/Administrator> is the administrator of the native domain. Apply the following syntax: DomainName/Username.
   - To set a list of users as administrators, add: `scanners.platform.auth.web.admin-users`. For example:

     ```
     scanners.platform.auth.web.admin-users=DomainName1/Username1;DomainName2/
     Username2;DomainName3/Username3
     ```

- To set a user group as administrators, add: `scanners.platform.auth.web.admin-groups=Native/group1`

  Note: Where <Native/group1> is the user group 1 from the native domain. Apply the following syntax: DomainName/GroupName.

- To set a list of groups as administrators, add: `scanners.platform.auth.web.admin-groups`. For example:

  ```
  scanners.platform.auth.web.admin-groups=DomainName1/Group1;DomainName2/Group2;DomainName3/Group3
  ```

3. Run the following commands to recycle the MetaDex server:

   ```
   INFA_HOME/services/CatalogService/AdvancedScannersApplication/app/sh server.sh stop
   INFA_HOME/services/CatalogService/AdvancedScannersApplication/app/sh server.sh start
   ```

Data upload

To run jobs, you must upload data to the MetaDex server.

You perform jobs on the server files, local machines are not available for jobs. To upload data to the server, you can:

- Use FTP
- Mount data as shared storage
- Upload through a web browser

Related Topics:

- "File management" on page 70

Enabling data upload

You can configure the ${INFA_HOME}/AdvScannersWorkspace to allow the users to upload data with a browser.

▶ Add or override a workspace by adding the following entry to the scanners.properties file: `scanners.properties.allowedDataLocations=/data/scanners/inputs,/workspace`

Uploading data

You can upload data with a browser in the MetaDex tool.

1. In the MetaDex tool, go to the **Administration** tab.
2. From the menu on the left, select **File Management**.

   The **File Management** page appears.
3. Click **Upload files**.

   Upload files window appears.

4. Click **Select files**.

   Open window appears.

5. Select the files you want to upload and click **Open**.

   Upload Files window appears.

6. Click **Upload**.

Content Security Policy

Content Security Policy (CSP) is a standard that prevents code injection attacks, such as cross-site scripting.

A website declares approved origins of content that a browser can load to display website content.

MetaDex has the following default rules: `scanners.web.security.contentPolicy = default-src 'self'; script-src 'self'; style-src 'self' 'unsafe-inline'; img-src 'self' data:; font-src 'self' data:; object-src 'none' ; trusted-types angular angular#unsafe-bypass; require-trusted-types-for 'script'; frame-ancestors 'none'`

You can optimize the policy with the `scanners.web.security.contentPolicy` server property.

Enrichment Migration Utility

The Enrichment Migration Utility is a utility enabled through the MetaDex tool. You can use the Enrichment Migration Utility to migrate enrichments between scanners of the same type. For example, you can migrate enrichments from the Salesforce scanner V1 to the Salesforce MetaDex scanner V2.

Use the Enrichment Migration Utility to migrate enrichments between scanners in the following situations:

- You changed the scanner name and need to scan the same metadata again.
- You started to use a newer version of a given scanner.
- You need to migrate enrichments between two instances of Enterprise Data Catalog, from one domain to another. For example, from "production" to "non-production".

You can migrate enriched metadata between test, staging, and production environments.

You can also use the utility to perform the following tasks:

- Back up enrichments of any resource.
- Copy enrichments from one environment to another using an exported backup.
- Restore enrichments using a previously exported backup.

The Enrichment Migration Utility works with the following enrichments:

- Business titles and out-of-the-box attributes like descriptions and business term associations.
- Business glossary associations.
- Certifications, including comments and tags.
- Any custom attribute values.
- User assignment data. For example, the data on the data owner, data steward and followers.

- User ratings and comments.
- Questions and answers.
- Curated data domains.

Configuring the Enrichment Migration Utility

When you install the Enrichment Migration Utility, it becomes available in the MetaDex tool.

1. Go to `${INFA_HOME}/AdvScannersWorkspace/etc` and create or open the external-tools-allow-list.properties file.
2. Add the following line to the external-tools-allow-list.properties file: `<Informatica installation directory>/services/CatalogService/AdvancedScannersApplication/app/enrichment-utility.sh`

   Note: Use the direct path to the installation directory instead of the INFA_HOME environment variable.
3. Save the file.

   The enrichment-utility.sh script can now operate as an external tool.
4. Restart the MetaDex server.
5. In the **MetaDex** tool navigate to **Administration** > **External Tools** > **Add external tool** and perform the following actions:
   a. In the **Name** field, enter `EDC Enrichment Migration` as the name of the tool.
   b. In the **Tool id** field, enter: `metadex-emu`
   c. In the **Command** field, enter the following path: `<Informatica installation directory>/services/CatalogService/AdvancedScannersApplication/app/enrichment-utility.sh`
   d. In the **Working directory** field, enter the following path: `<Informatica installation directory>/AdvScannersWorkspace`
   e. In the **Log layout** field, enter the following data: `%d{HH:mm:ss.SSS} %p %m`
   f. In the **Config schema** field, enter the content of the migration-utility-schema.json file.

      The file is available in the following location: `${INFA_HOME}/services/CatalogService/AdvancedScannersApplication/app/utils/enrichment-migration-utility/schemas/migration-utility-schema.json`

Copying enrichments

Copy enrichments between resources of the same type. To copy an enrichment from one resource to another, perform the following actions:

1. In the MetaDex tool, go to the **More** menu (⋮) and select **Create configuration** > **EDC Enrichment Migration**.
2. Set the **Write to Target** parameter to **True**.
3. Provide the source and target resource names.
4. Save and run the configuration.

Backing up enrichments

You can use the Enrichment Migration Utility to back up enrichments. To back up an enrichment of any resource, perform the following actions:

1. In the MetaDex tool, go to the **More** menu (⋮) and select **Create configuration** > **EDC Enrichment Migration**.
2. Set the **Write to Target** parameter to **False**.
3. Set the **Export Assets to Directory** parameter to **True**.
4. Provide the backup directory.
5. Save and run the configuration.

To restore an enrichment, use a previously exported backup.

APPENDIX A

Metadata Extractor

This appendix includes the following topics:

- Metadata Extractor overview, 85
- Metadata Extractor reference, 86

Metadata Extractor overview

Metadata Extractor is a standalone console tool that connects to a client database and unloads tables into .csv files. You can later use those files as metadata repositories in MetaDex scanner jobs instead of connecting directly to the client database.

Metadata Extractor is included in the MetaDex distribution package.

Depending on your operating system, run the `extractMetadata.bat` or `extractMetadata.sh` script located in the `metadata-extractor.zip`.

The `extractMetadata` script performs the following actions:

- Downloads a client database catalog.
- Compresses the .csv files into a single .zip archive file.

Use the archive to emulate a client database by configuring `repoType=CATALOG` in the db.properties file.

The `lib` folder includes all used libraries, as well as libraries that you might need.

Note: Place the JDBC drivers in the `lib` folder on the metadata-extractor classpath. For more information, see "JDBC drivers" on page 48.

Metadata Extractor reference

Use the parameters provided in this section for extracting metadata from databases using the Metadata Extractor tool.

Metadata Extractor parameters

The following table describes the Metadata Extractor parameters:

Parameter	Description
--jdbcUrl, -url	Required. Database URL, for example `-url jdbc:oracle:thin:@oracle01.compactbi.local:1521:cspl`
--username, -u	Required. User name.
--password, -p	Required. Password; can be encrypted with MetaDex encryption tools.
--schema, -s	Required. Schema pattern to unload. Supported wildcard characters: - One character: _ - Any number of characters: % You can separate patterns with a pipe (|). For example: - -s mySchema - -s mySche_a - -s my%Schema - -s "dbo|sys"
--sqlserver-2005, -old-mssql	Extracts Microsoft SQL Server 2005.
--netezza-without-schemas-support, -no-schemas-netezza	Extracts Netezza without schemas support.
--output, -o	Overrides the default output file name.

Required permissions

The following table lists the permissions required by Metadata Extractor:

Database type	Permissions
UDB DB2	- select on syscat.COLUMNS - select on syscat.FUNCTIONS - select on syscat.NICKNAMES - select on syscat.PROCEDURES - select on syscat.PROCPARMS - select on syscat.TABLES - select on syscat.USEROPTIONS - select on syscat.VIEWS
Greenplum	- select on pg_catalog.PG_ATTRIBUTE - select on pg_catalog.PG_CLASS - select on pg_catalog.PG_CONSTRAINT - select on pg_catalog.PG_LANGUAGE - select on pg_catalog.PG_NAMESPACE - select on pg_catalog.PG_PROC - select on pg_catalog.PG_TYPE - select on pg_catalog.PG_VIEWS
Hive	- select on sys.COLUMNS_V2 - select on sys.DBS - select on sys.PARTITION_KEYS - select on sys.SDS
Netezza No Schemas	- select on _V_OBJECTS - select on _V_PROCEDURE - select on _V_SCHEMA - select on _V_SEQUENCE - select on _V_SYNONYM - select on _V_VIEW - select on _V_JDBC_COLUMNS1
Netezza Schema Support	- select on _V_OBJECTS - select on _V_PROCEDURE - select on _V_SCHEMA - select on _V_SEQUENCE - select on _V_SYNONYM - select on _V_VIEW - select on _V_RELATION_COLUMN

Database type	Permissions
Oracle	- select on SYS.DBA_ARGUMENTS - select on SYS.DBA_COL_COMMENTS - select on SYS.DBA_COLL_TYPES - select on SYS.DBA_CONS_COLUMNS - select on SYS.DBA_CONSTRAINTS - select on SYS.DBA_DB_LINKS - select on SYS.DBA_DIRECTORIES - select on SYS.DBA_EXTERNAL_LOCATIONS - select on SYS.DBA_EXTERNAL_TABLES - select on SYS.DBA_MVIEWS - select on SYS.DBA_OBJECTS - select on SYS.DBA_OPBINDINGS - select on SYS.DBA_SEQUENCES - select on SYS.DBA_SOURCE - select on SYS.DBA_SYNONYMS - select on SYS.DBA_TAB_COLUMNS - select on SYS.DBA_TAB_COMMENTS - select on SYS.DBA_TABLES - select on SYS.DBA_TRIGGERS - select on SYS.DBA_TRIGGERS - select on SYS.DBA_USERS - select on SYS.DBA_VIEWS - select on SYS.NLS_SESSION_PARAMETERS
Redshift	- select on pg_catalog.PG_ATTRIBUTE - select on pg_catalog.PG_CLASS - select on pg_catalog.PG_CONSTRAINT - select on pg_catalog.PG_LANGUAGE - select on pg_catalog.PG_NAMESPACE - select on pg_catalog.PG_PROC - select on pg_catalog.PG_TYPE - select on pg_catalog.PG_VIEWS - select on information_schema.TABLES - select on pg_catalog.PG_CLASS_INFO - select on pg_catalog.PG_DESCRIPTION - select on pg_catalog.PG_TABLES
SAP HANA	- select on SYS.SYNONYMS - select on SYS.TABLE_COLUMNS - select on SYS.VIEW_COLUMNS
Snowflake	- select on information_schema.EXTERNAL_TABLES - select on information_schema.FUNCTIONS - select on information_schema.PIPES - select on information_schema.PROCEDURES - select on information_schema.SCHEMATA - select on information_schema.SEQUENCES - select on information_schema.STAGES - select on information_schema.VIEWS - show objects - show columns - show primary keys - show imported keys - show streams - show materialized views - show tasks

Database type	Permissions
Microsoft SQL Server	- select on sys.ALL_COLUMNS - select on sys.ALL_OBJECTS - select on sys.ALL_PARAMETERS - select on sys.DATABASE_PRINCIPALS - select on sys.FOREIGN_KEY_COLUMNS - select on sys.INDEXES - select on sys.INDEX_COLUMNS - select on sys.SCHEMAS - select on sys.SQL_MODULES - select on sys.SYNONYMS - select on sys.TYPES - select on sys.TABLE_TYPES
Microsoft SQL Server 2005	- select on sys.ALL_COLUMNS - select on sys.ALL_OBJECTS - select on sys.ALL_PARAMETERS - select on sys.DATABASE_PRINCIPALS - select on sys.FOREIGN_KEY_COLUMNS - select on sys.INDEXES - select on sys.INDEX_COLUMNS - select on sys.SCHEMAS - select on sys.SQL_MODULES - select on sys.SYNONYMS - select on sys.TYPES
Sybase	- select on dbo.syscolumns - select on dbo.syscomments - select on dbo.sysobjects - select on dbo.sysservers - select on dbo.systypes - select on dbo.sysusers
Teradata	- select on dbc.COLUMNSV - select on dbc.TABLESV - select on dbc.TABLETEXTV

JDBC URL examples

The following table lists the JDBC URL examples:

Database type	URL example
DB2 iSeries, LUW, zOS	jdbc:db2://127.0.0.1:50000/SAMPLE
Hive	jdbc:hive2://hive01:1234/database
Netezza	jdbc:netezza://netezza01:5480/DBA
Oracle	jdbc:oracle:thin:@oraclehost:1521:sid
Microsoft SQL Server	jdbc:sqlserver://mssql01:51433;databaseName=prod
Teradata	jdbc:teradata://127.0.0.1/1025/Teradata01

Note: You can use the extracted metadata in MetaDex as an offline catalog.

APPENDIX B

PL/SQL calls

View the explanations of the required structure of the file containing the information about the calls of PL/SQL procedures or functions processed by the MetaDex server.

Syntax convention

A CSV file can contain any number of procedure calls. Each procedure call is stored in one line of the file.

A single procedure call is defined in one line, for example:

```
"callIdentifier", "procedureIdentifier (argumentList?)", "optionalComment"
```

where:

callIdentifier	Required. A string identifying a particular procedure call. Use it as a part of object name when loaded as a lineage into Common Metadata repository.
procedureIdentifier	Required. Identifier of the database-stored procedure or function as recognized and called in the database. Syntax depends on the database vendor. For example, in Oracle: `schema.packageName.procedureName@dbLinkName`, where specifying `schema`, `packageName`, or `@dbLinkName` are optional. If the procedure owner is not specified, it is assumed to be the value inherited from the `Context` value.
argumentList	Optional. The list of the actual parameters in the procedure. If the list is empty, the brackets "()" are required.
optionalComment	Optional. The comment is placed in Description field of the object representing the lineage extracted from this procedure call.

Note: Input file must be a valid .csv file.

You can omit the quotation marks if the text does not contain comma nor NEW_LINE character.

You can escape quotation marks with quotation marks.

PL/SQL calls file example:

```
CallId,Call,Comment
calling.from.populateTransactionFACTs.ksh, HD_SCHEMA.TRN_FACT('Select datetime
""now"" from dual'), "Initial load for Transaction
 Fact and Auxiliary Fact table, invoked from .ksh"

accountSCD.sh, "DIM_SCHEMA.DO_SCD2('ODS.ACCOUNT','ACCOUNT')", updating ACCOUNT
conformed dimension table based on ODS.ACCOUNT
 table using SCD 2 approach. Invoked from .sh
```

The example contains two procedure calls. Each procedure can create multiple data flows, resulting with separate lineages.

View the pieces of lineage from the first row:

- `calling.from.populateTransactionFACTs.ksh.23` means that an identified lineage-carrying code starts in line 23.
- `calling.from.populateTransactionFACTs.ksh.167` means that an identified lineage-carrying code starts in line 167.

Names of the generated lineage objects depend on the defined naming pattern.

APPENDIX C

SAS code hints

View hints for SAS code development analyzed by MetaDex scanners with minimum required annotations.

Essenial terms

Annotation	An annotation is an additional remark in source code. A special comment that does not impact the functionality of the original code. Use an annotation to provide the MetaDex server with hints to infer valid lineage. SAS-Annotations help tracking execution-time values of SAS-macrovariables. Annotations begin with /*@ and end with @*/.

MetaDex scanners SAS-engine general information

You receive a warning each time you use a value of unknown-value-macrovariable for code generation. It can always affect the lineage.

MetaDex scanners SAS-engine has similar architecture to SAS-engine. It generates SAS-steps by solving Macro language.

Function calls, Macro Statements, and Macro Variables are supported.

The MetaDex server performs code analysis, meaning it does not know the dynamic values such as: current date, data fetched from databases, or data sets. If a SAS program uses such value in data transformation, you need to provide hints to calculate lineage.

Current version of MetaDex can process the macro loops and macro conditional statements with restrictions. MetaDex-macroprocessor generates code of one loop and all statements in the `%if` and `%else` block.

For complex macros, which output does not affect a lineage, you can define a dummy macro for lineage calculation.

Call Symput function and %LET annotation example

In the following example, you set the value of a macro variable:

`today_`*`DDMMYY`*

`month_ago_`*`DDMMYY`*

Then you create a report named `REPORT_DDMMYY`. The name ensures that one lineage extraction shows every case from the past and future.

Ensure that the MetaDex server gets the correct value of the Macro variable *today_DDMMYY* by adding an annotation with the generic value of Macro variables set by the `CALL SYMPUT(X)` function. The annotation is a hint to interpret the current code. You can provide any code understandable for the Macro Processor. Usually, the `%LET` statement is enough.

```
... data _null_;
 today_DDMMYY = some_calculation(today());
```

```
 month_ago_DDMMYY = some_calculation(today());
 call symput('today_DDMMYY',today_DDMMYY);
 call symput('month_ago_DDMMYY',month_ago_DDMMYY);
run; /*@ %let today_DDMMYY=DDMMYY; %let month_ago_DDMMYY=DDMMYY; @*/

data REPORT_&today_DDMMYY;
    set REPORT_&month_ago_DDMMYY;
... more report calculations;
run;
...
```

Handling complex code that does not affect lineage

For transformations that do not have impact on the lineage generation and cause conversion errors, use an `IGNORE` annotation. The `IGNORE` annotation is a `BLOCK-COMMENT`. It prevents this code from analysis.

```
...
/*@ ignore:begin @*/
data _null_;
...Complex Calculation Which Does Not Have Impact On Lineage Generation...
run;
/*@ ignore:end @*/
...
```

APPENDIX D

Scanners-cli utility

You can use the scanners-cli utility to manage the MetaDex server from the command line.

Run the scanners-cli utility from the following directory: `Informatica installation directory/services/CatalogService/AdvancedScannersApplication/app/bin`

You can run the following commands:

- assign-role
- blueprints new
- blueprints parameters set
- config set
- config show
- configurations parameters set
- conversionpaths
- create project
- export blueprint
- export project
- export variableset
- import blueprint
- import project
- import variableset
- processings abort-all
- processings abort-by-id
- processings list
- processings remove
- run
- show license
- show project
- show projects
- status

Assign-role

Run the assign-role command to assign roles to users. Specify the roles, users, and projects in the command options.

The assign-role command uses the following syntax:

```
assign-role [-h] [--json] [-p=<password>] --role=<roleString> [-s=<url>] [-
u=<user>] --projects=<projectname>[,<projectname>...] [--projects=<projectname>
[,<projectname>...]]... --users=<username>[,<username>...] [--
users=<username>[,<username>...]]
```

The following table describes command properties:

Property	Description
-h, --help	Displays help and exits.
--json	Enables JSON file output format.
-p, --password=<password>	Password. You can: - Encrypt the password with the encrypt script - Specify the password with the SCANNERS_PASSWORD environment variable
--role=<roleString>	The role that you assign to the user in a given project.
-s, --url=<url>	URL of the Enterprise Data Catalog web server. For example. http://10. 1.1.122:8090. You can specify the URL with the SCANNERS_URL environment variable.
-u, --user=<user>	User name. Use the SCANNERS_USER environment variable to specify users.
-projects=<projectname>[,<projectname>...]	Comma-separated names of the projects to which you assign the users to, for example: projectA,projectB,projectC
--users, --usernames=<username>[,<username>...]	Comma-separated names of the users whom you assign the roles. For example: userA,userB,userC

Blueprints new

Run the blueprints-new command to create a configuration with a unique name.

Th blueprints-new command uses the following syntax:

```
blueprints new [-h] [--json] [-p=<password>] --project=<projectName> [-s=<url>]
[-u=<user>] NAME CONVERSIONPATH [DESCRIPTION]
```

The following table describes command properties:

Property	Description
-h, --help	Displays help and exits.
--json	Enables JSON file output format.
-p, --password=<password>	Password. You can: - Encrypt the password with the encrypt script - Specify the password with the SCANNERS_PASSWORD environment variable
--project=<projectName>	Required. The name of the project.
-s, --url=<url>	Required. URL of the Enterprise Data Catalog web server. For example. http://10.1.1.122:8090. You can specify the URL with the SCANNERS_URL environment variable.
-u, --user=<user>	User name. Use the SCANNERS_USER environment variable to specify users.
NAME	Required. The name of the configuration.
CONVERSIONPATH	Required. The conversion path ID.
DESCRIPTION	Description of the configuration.

Blueprints parameters set

Run the blueprints parameters set command to set configuration parameters.

The blueprints parameters set command uses the following syntax:

```
blueprints parameters set [-h] [--json] --blueprint=<blueprintName> [-
p=<password>] --project=<projectName> [-s=<url>] [-u=<user>] [KEY=VALUE...]
```

The following table describes command properties:

Property	Description
-h, --help	Displays help and exits.
--json	Enables JSON file output format.
--blueprint=<blueprintName>	Required. The name of the configuration.
-p, --password=<password>	Password. You can: - Encrypt the password with the encrypt script - Specify the password with the SCANNERS_PASSWORD environment variable
--project=<projectName>	Required. The name of the project.
-s, --url=<url>	Required. URL of the Enterprise Data Catalog web server. For example. http://10.1.1.122:8090. You can specify the URL with the SCANNERS_URL environment variable.

Property	Description
-u, --user=<user>	User name. Use the SCANNERS_USER environment variable to specify users.
[KEY=VALUE]	The value of the parameter.

Config set

Run the config set command to set the user name, URL of the Enterprise Data Catalog web server, and an encrypted password for every command you use in the scanners-cli utility.

The config set command uses the following syntax:

```
config set user=sampleUserName
```

The following table describes command properties:

Property	Description
--path	Optional. Allows you to use a non-standard path for the configuration.

Config show

Run the config show command to view the KEY=VALUE pairs in the configuration file. By default, the configuration file is the scanners-cli-config file in your home catalog.

The config show command uses the following syntax:

```
config show [--path]
```

The following table describes command properties:

Property	Description
--path	Optional. Allows you to use a non-standard path for the configuration.

Configurations parameters set

Run the configurations parameters set command to update configuration properties.

The configurations parameters set command uses the following syntax:

```
scanners-cli.bat configurations parameters set --project <projectName> --
configuration=<configurationName> -d <security_domain_name> -u <user> -p
<password> -s <url> <Configuration key-value list>
```

The following table describes command properties:

Property	Description
--project=<projectName>	Required. The name of the project.
--configuration=<configurationName>	Required. The name of the configuration.
-d, --domain=<security_domain_name>	Required. The name of the security domain.
-u, --user=<user>	User name. Specify the user in one of the following ways: - Encrypt the user name with the encrypt script. - Specify the user name with the SCANNERS_USER environment variable.
-p, --password=<password>	Password. Specify the password in one of the following ways: - Encrypt the password with the encrypt script. - Specify the password with the SCANNERS_PASSWORD environment variable.
-s, --url=<url>	URL of the Enterprise Data Catalog web server. For example. http://10. 1.1.122:8090. Specify the URL in one of the following ways: - Encrypt the URL with the encrypt script. - Specify the URL with the SCANNERS_URL environment variable.
<Configuration key-value list>	Required. Space-separated list of parameters that you want to update. The parameters are listed in the following format: <key>=<value>. For example: inputRoot=/path/root plSqlInputScripts=/path/to/script

For example, if you want to change the object include filter of the 'TEST_PROJECT' with the 'TEST' configuration to 'HOSTB.DBB.SCHEMAB', enter the following command: `./scanners-cli.sh configurations parameters set --project TEST_PROJECT --configuration TEST --user admin --password admin --url http://10. 1.1.122:8090 sql.objectIncludeFilter=HOSTB.DBB.SCHEMAB`

Where the user name is 'admin', the password is 'admin' and the URL is 'http://10. 1.1.122:8090'.

Conversionpaths

Run the conversionpaths command to view available conversion paths.

The conversionpaths command uses the following syntax:

```
conversionpaths [-h] [--json] [-p=<password>] [-s=<url>] [--show-
parameters=<conversionPathId>] [-u=<user>]
```

The following table describes command properties:

Property	Description
-h, --help	Displays help and exits.
--json	Enables JSON file output format.
-p, --password=<password>	Password. You can: - Encrypt the password with the encrypt script - Specify the password with the SCANNERS_PASSWORD environment variable
-s, --url=<url>	Required. URL of the Enterprise Data Catalog web server. For example. http://10. 1.1.122:8090. You can specify the URL with the SCANNERS_URL environment variable.
--show-parameters=<conversionPathId>	Show available parameters for a given conversion path.
-u, --user=<user>	User name. Use the SCANNERS_USER environment variable to specify users.

Create project

Run the create project command to create a project.

The create project command uses the following syntax:

```
create project [-h] [--json] [--code=CODE] [-p=<password>] [-s=<url>] [-
u=<user>] NAME [DESCRIPTION]
```

The following table describes command properties:

Property	Description
-h, --help	Displays help and exits.
--json	Enables JSON file output format.
--code=CODE	The code of the project. By default based on the name of the project.
-p, --password=<password>	Password. You can: - Encrypt the password with the encrypt script - Specify the password with the SCANNERS_PASSWORD environment variable
-s, --url=<url>	URL of the Enterprise Data Catalog web server. For example. http://10. 1.1.122:8090. You can specify the URL with the SCANNERS_URL environment variable.
-u, --user=<user>	User name. Use the SCANNERS_USER environment variable to specify users.

Property	Description
NAME	Required. The name of the project.
[DESCRIPTION]	The description of the project.

Export blueprint

Run the export blueprint command to export configurations.

The export blueprint command uses the following syntax:

```
export blueprint [-h] [--json] [--with-notifications] --
blueprint=<blueprintName> [-f=<file>] [-p=<password>] --project=<projectName> [-
s=<url>] [-u=<user>]
```

The following table describes command properties:

Property	Description
-h, --help	Displays help and exits.
--json	Enables JSON file output format.
--with-notifications	Adds the configuration of email notifications.
--blueprint=<blueprintName>	Required. The name of the configuration.
-f, --file=<file>	Relative or absolute file path to the file that you export.
-p, --password=<password>	Password. You can: - Encrypt the password with the encrypt script - Specify the password with the SCANNERS_PASSWORD environment variable
--project=<projectName>	Required. The name of the project.
-s, --url=<url>	URL of the Enterprise Data Catalog web server. For example. http://10. 1.1.122:8090. You can specify the URL with the SCANNERS_URL environment variable.
-u, --user=<user>	User name. Use the SCANNERS_USER environment variable to specify users.

Export project

Run the export project command to export projects.

The export project command uses the following syntax:

```
export project [-h] [--json] [--with-notifications] [--with-permissions] [-
f=<exportFile>] [-p=<password>] [-s=<url>] [-u=<user>] <ExportedProjects>
```

The following table describes command properties:

Property	Description
-h, --help	Displays help and exits.
--json	Enables JSON file output format.
--with-notifications	Adds the configuration of email notifications.
--with-permissions	Exports the project with existing user and group permissions.
-f, --file=<exportFile>	Name of the target ZIP archive.
-p, --password=<password>	Password. You can: - Encrypt the password with the encrypt script - Specify the password with the SCANNERS_PASSWORD environment variable
-s, --url=<url>	URL of the Enterprise Data Catalog web server. For example. http://10. 1.1.122:8090. You can specify the URL with the SCANNERS_URL environment variable.
-u, --user=<user>	User name. Use the SCANNERS_USER environment variable to specify users.
<ExportedProjects>	Required. A list of the projects packed into the ZIP archive.

Export variableset command

Run the export variableset command to export sets of variables.

The export variableset command uses the following syntax:

```
export variableset [-h] [--json] --blueprint=<blueprintName> [-f=<file>] [-
p=<password>] --project=<projectName> [-s=<url>] [-u=<user>] --
variableset=<variableSetName>
```

The following table describes command properties:

Property	Description
-h, --help	Displays help and exits.
--json	Enabless JSON file output format.
--blueprint=<blueprintName>	Required. The name of the blueprint.
-f, --file=<file>	Relative or absolute file path to the file that you export.

Property	Description
-p, --password=<password>	Password. You can: - Encrypt the password with the encrypt script - Specify the password with the SCANNERS_PASSWORD environment variable
--project=<projectName>	Required. The name of the project.
-s, --url=<url>	URL of the Enterprise Data Catalog web server. For example. http://10. 1.1.122:8090. You can specify the URL with the SCANNERS_URL environment variable.
-u, --user=<user>	User name. Use the SCANNERS_USER environment variable to specify users.
--variableset=<variableSetName>	Required. The name of the variable set.

Import blueprint

Run the import blueprint command to import configurations.
The import blueprint command uses the following syntax:

```
import blueprint [-h] [--json] [--override] [--with-notifications] -
f=<sourceFile> [--name=<blueprintName>] [-p=<password>] --project=<projectName>
[-s=<url>] [--tool=<toolId>] [-u=<user>]
```

The following table describes command properties:

Property	Description
-h, --help	Displays help and exits.
--json	Enables JSON file output format.
--override	Overwrites the configuration, if a configuration with given name already exists.
--with-notifications	Adds the configuration of email notifications.
-f, --file=<sourceFile>	A path to the properties file that contains the configuration.
--name=<blueprintName>	The configuration name. If you do not provide the name, the properties file name is the name of the configuration.
-p, --password=<password>	Password. You can: - Encrypt the password with the encrypt script - Specify the password with the SCANNERS_PASSWORD environment variable
--project=<projectName>	Required. The name of the project.
-s, --url=<url>	URL of the Enterprise Data Catalog web server. For example. http://10. 1.1.122:8090. You can specify the URL with the SCANNERS_URL environment variable.

Property	Description
--tool=<toolId>	The name of the configuration tool. The default name is MetaDex.
-u, --user=<user>	User name. Use the SCANNERS_USER environment variable to specify users.

Import project

Run the import project command to import projects.

The import project command uses the following syntax:

```
import project [-h] [--json] [--with-notifications] [--with-permissions] -
f=<FileName> [-p=<password>] [-s=<url>] [-u=<user>] <ExportedProjects>
```

The following table describes command properties:

Property	Description
-h, --help	Displays help and exits.
--json	Enables JSON file output format.
--with-notifications	Adds the configuration of email notifications.
--with-permissions	Imports user and group permissions if they are in the given file.
-f, --file=<FileName>	Required. A ZIP file with exported configurations. If you do not specify the projectName parameter, the file name is the project name.
-p, --password=<password>	Password. You can: - Encrypt the password with the encrypt script - Specify the password with the SCANNERS_PASSWORD environment variable
-s, --url=<url>	URL of the Enterprise Data Catalog web server. For example. http://10. 1.1.122:8090. You can specify the URL with the SCANNERS_URL environment variable.
-u, --user=<user>	User name. Use the SCANNERS_USER environment variable to specify users.

Import variableset

Run the import variableset command to import sets of variables.

The import variableset command uses the following syntax:

```
import variableset [-h] [--json] [--override] -f=<filePath> [-n=<vsName>] [-
p=<password>] [-s=<url>] [-u=<user>] <BlueprintLocation>
```

The following table describes command properties:

Property	Description
-h, --help	Displays help and exits.
--json	Enables JSON file output format.
--override	Overwrites the variable set if a variable set with given name already exists.
-f, --file=<filePath>	Required. A path to the CSV file that contains the variable set.
-n, --name=<vsName>	The variable set name. If you do not provide the name, the CSV file name is the name of the variable set.
-p, --password=<password>	Password. You can: - Encrypt the password with the encrypt script - Specify the password with the SCANNERS_PASSWORD environment variable
-s, --url=<url>	URL of the Enterprise Data Catalog web server. For example. http://10. 1.1.122:8090. You can specify the URL with the SCANNERS_URL environment variable.
-u, --user=<user>	User name. Use the SCANNERS_USER environment variable to specify users.
<BlueprintLocation>	Required. A project-qualified location. You import the variable set for the specified configuration. For example: `project1/config1, "project 2/config2" or project1/config2/overridden1`

Processings abort-all

Run the processings abort-all command to stop all jobs.

The processings abort-all command uses the following syntax:

```
processings abort-all [-h] [--json] [-p=<password>] [-s=<url>] [-u=<user>]
```

The following table describes command properties:

Property	Description
-h, --help	Displays help and exits.
--json	Enables JSON file output format.
-p, --password=<password>	Password. You can: - Encrypt the password with the encrypt script - Specify the password with the SCANNERS_PASSWORD environment variable

Property	Description
-s, --url=<url>	Required. URL of the Enterprise Data Catalog web server. For example. http://10. 1.1.122:8090. You can specify the URL with the SCANNERS_URL environment variable.
-u, --user=<user>	User name. Use the SCANNERS_USER environment variable to specify users.

Processings abort-by-id

Run the processings abort-by-id command to stop jobs by their job ID.

The processings abort-by-id command uses the following syntax:

```
processings abort-by-id [-h] [--json] [-p=<password>] [-s=<url>] [-u=<user>]
<Runnables List>
```

The following table describes command properties:

Property	Description
-h, --help	Displays help and exits.
--json	Enables JSON file output format.
-p, --password=<password>	Password. You can: - Encrypt the password with the encrypt script - Specify the password with the SCANNERS_PASSWORD environment variable
-s, --url=<url>	URL of the Enterprise Data Catalog web server. For example. http://10. 1.1.122:8090. You can specify the URL with the SCANNERS_URL environment variable.
-u, --user=<user>	User name. Use the SCANNERS_USER environment variable to specify users.
<Runnables List>	Required. Space-separated, project-qualified list of objects you want to stop. For example: `project1/config1 "project 2/config2/variableSet1" project1/config2/overridden1/vs1`

Processings list

Run the processings list command to list jobs.

The processings list command uses the following syntax:

```
processings list [-h] [--json] [-p=<password>] [-s=<url>] [-u=<user>]
```

The following table describes command properties:

Property	Description
-h, --help	Displays help and exits.
--json	Enables JSON file output format.
-p, --password=<password>	Password. You can: - Encrypt the password with the encrypt script - Specify the password with the SCANNERS_PASSWORD environment variable
-s, --url=<url>	Required. URL of the Enterprise Data Catalog web server. For example. http://10. 1.1.122:8090. You can specify the URL with the SCANNERS_URL environment variable.
-u, --user=<user>	User name. Use the SCANNERS_USER environment variable to specify users.

Processings remove

Run the processings remove command to remove jobs.

The processings remove command uses the following syntax:

```
processings remove [-Dh] [--json] [-o=<olderThen>] [-p=<password>] [-s=<url>] [-u=<user>] <Runnables List>
```

The following table describes command properties:

Property	Description
-h, --help	Displays help and exits.
--json	Enables JSON file output format.
-o, --olderThan=<olderThan>	Removes jobs older than the number you specify. For example, if you enter 3, the fourth, fifth, and later jobs are removed.
-p, --password=<password>	Password. You can: - Encrypt the password with the encrypt script - Specify the password with the SCANNERS_PASSWORD environment variable
-s, --url=<url>	URL of the Enterprise Data Catalog web server. For example. http://10. 1.1.122:8090. You can specify the URL with the SCANNERS_URL environment variable.
-u, --user=<user>	User name. Use the SCANNERS_USER environment variable to specify users.
<Runnables List>	Required. Space-separated, fully qualified list of objects that you want to remove. For example: `project1/config1 "project 2/config2/variableSet1" project1/config2/overridden1/vs1`

Run

Run the run command to run scheduled objects.

The run command uses the following syntax:

```
run [-hvw] [--ignore-hazardous-values] [--json] [-p=<password> [-s=<url>] [-
u=<user>] <Runnables List>
```

The following table describes command properties:

Property	Description
-h, --help	Displays help and exits.
-v, --verbose	Shows more details about the results of running objects. You can use it when results of the run are not obvious.
-w, --wait	Waits until the scheduled job is finished.
--ignore-hazardous-values	Runs despite hazardous parameter values that can lead to serious issues.
--json	Enables JSON file output format.
-p, --password=<password>	Password. You can: - Encrypt the password with the encrypt script - Specify the password with the SCANNERS_PASSWORD environment variable
-s, --url=<url>	URL of the Enterprise Data Catalog web server. For example. http://10. 1.1.122:8090. You can specify the URL with the SCANNERS_URL environment variable.
-u, --user=<user>	User name. Use the SCANNERS_USER environment variable to specify users.
<Runnables List>	Required. Space-separated, fully qualified list of objects that you want to schedule. For example: `project1/config1 "project 2/config2/variableSet1" project1/config2/overridden1/vs1` **Note:** Objects you scheduled together may not execute immediately. That depends on the number of allowed concurrent jobs that you set during the server configuration.

Show license

Run the show license command to view license details, such as the expiration date or available modules. The show license command uses the following syntax:

```
show license [-h] [--json] [-p=<password>] [-s=<url>] [-u=<user>]
```

The following table describes command properties:

Property	Description
-h, --help	Displays help and exits.
--json	Enables JSON file output format.
-p, --password=<password>	Password. You can: - Encrypt the password with the encrypt script - Specify the password with the SCANNERS_PASSWORD environment variable
-s, --url=<url>	Required. URL of the Enterprise Data Catalog web server. For example. http://10.1.1.122:8090. You can specify the URL with the SCANNERS_URL environment variable.
-u, --user=<user>	User name. Use the SCANNERS_USER environment variable to specify users.

Show project

Run the show project command to view the configuration and variable sets of the project.
The show project command uses the following syntax:

```
show project [-h] [--json] [-p=<password>] [--project=NAME] [-s=<url>] [-
u=<user>] [NAME]
```

The following table describes command properties:

Property	Description
-h, --help	Displays help and exits.
--json	Enables JSON file output format.
-p, --password=<password>	Password. You can: - Encrypt the password with the encrypt script - Specify the password with the SCANNERS_PASSWORD environment variable
--project=NAME	The name of the project.
-s, --url=<url>	Required. URL of the Enterprise Data Catalog web server. For example. http://10.1.1.122:8090. You can specify the URL with the SCANNERS_URL environment variable.
-u, --user=<user>	User name. Use the SCANNERS_USER environment variable to specify users.

Show projects

Run the show projects command to view projects.
The show projects command uses the following syntax:

```
show projects [-h] [--json] [-p=<password>] [-s=<url>] [-u=<user>]
```

The following table describes command properties:

Property	Description
-h, --help	Displays help and exits.
--json	Enables JSON file output format.
-p, --password=<password>	Password. You can: - Encrypt the password with the encrypt script - Specify the password with the SCANNERS_PASSWORD environment variable
-s, --url=<url>	Required. URL of the Enterprise Data Catalog web server. For example. http://10. 1.1.122:8090. You can specify the URL with the SCANNERS_URL environment variable.
-u, --user=<user>	User name. Use the SCANNERS_USER environment variable to specify users.

Status command

This section describes the command details.

The status command shows the status of a job you run.

See the following usage example: `status [-h] [--json] [-p=<password>] [-s=<url>] [-u=<user>] processingId`

The following table describes command properties:

Property	Description
-h, --help	Displays help and exits.
--json	Enables JSON file output format.
-p, --password=<password>	Password. You can: - Encrypt the password with the encrypt script - Specify the password with the SCANNERS_PASSWORD environment variable
-s, --url=<url>	URL of the Enterprise Data Catalog web server. For example. http://10. 1.1.122:8090. You can specify the URL with the SCANNERS_URL environment variable.
-u, --user=<user>	User name. Use the SCANNERS_USER environment variable to specify users.
processingId	Required. ID of the job that you run.

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